Acute Alteration in Respiratory Functions



BY: Dr. Wael S. Mekky

ACUTE RESPIRATORY FAILURE

Respiratory Failure

- A condition that occurs when respiratory system is unable to adequately exchange Oxygen and Carbon Dioxide between environment and tissues of the body.
- Respiratory failure is a sudden and life-threatening deterioration of the gas exchange function of the lung; oxygenation and carbon dioxide elimination.

Respiratory Failure

- Acute respiratory failure (ARF) is defined as a fall in arterial oxygen tension (PaO_2) to less than 60 mm Hg (hypoxemia) or/and a rise in arterial carbon dioxide tension ($PaCO_2$) to greater than 50 mm Hg (hypercapnia), with an arterial pH of less than 7.35.

Respiratory Failure

- PaO_2 < 60 mmHg
- $PaCO_2$ > 50 mmHg

Respiratory Failure

- It is important to distinguish between ARF and chronic respiratory failure.
- Chronic Respiratory Failure is defined as deterioration in the gas exchange function of the lung that has developed gradually or has persisted for a long period after an episode of ARF.

Respiratory Failure

- Two causes of chronic respiratory failure are COPD and neuromuscular diseases. Patients with these disorders develop a tolerance to the gradually worsening hypoxemia and hypercapnia.

Types of Respiratory Failure

- ***Type I :***

  Oxygenation Failure (Hypoxemic)

- ***Type II :***

  Ventilation Failure (Hypercapnic)

Types of Acute Respiratory Failure

1. Acute hypoxemic respiratory failure (Type I):

Hypoxia without hypercapnia, the CO_2 level may be normal or low. This is the most common form of respiratory failure, and it can be associated with all acute diseases of the lung. Some examples of type I respiratory failure are cardiogenic or noncardiogenic pulmonary edema, and pneumonia.

Cardiogenic Pulmonary Edema:

If a heart problem causes the pulmonary edema, it's called cardiogenic pulmonary edema.

Noncardiogenic Pulmonary Edema:

If pulmonary edema is not heart related, it's called noncardiogenic pulmonary edema.

2. Acute hypercapnic respiratory failure (Type II):

- Characterized by a $PaCO_2$ higher than 50 mmHg.
- The pH depends on the level of bicarbonate, which, in turn, is dependent on the duration of hypercapnia.
- It is due to inadequate air flow in the alveoli of the lungs that causes a buildup of carbon dioxide that has been generated by the body.

2. Acute hypercapnic respiratory failure (Type II):

The underlying etiologies are:

- ✓ Reduced breathing effort (in the fatigued patient).
- ✓ Increased resistance to breathing (such as in asthma and COPD.
- ✓ Increase in the area of the lung that is not available for gas exchange (such as in emphysema).

Causes Of Respiratory Failure

- The causes of acute respiratory failure can be grouped according to the primary abnormality and the individual components of the respiratory system (eg, CNS (medulla), peripheral nervous system, respiratory muscles, chest wall, airways, and alveoli) as **the next:**

Types of Acute Respiratory Failure

1- Airways Obstruction:

✓ Severe airway obstruction is a common cause of acute and chronic hypercapnia.

✓ Examples of upper-airway disorders are acute epiglottitis and tumors involving the trachea, bronchiolitis; lower-airway disorders include COPD, asthma, and cystic fibrosis.

Types of Acute Respiratory Failure

2. Alveolar and pleural disease:

- Diseases of the alveoli are characterized by diffuse alveolar filling, frequently resulting in hypoxemic respiratory failure, although hypercapnia may complicate the clinical picture.
- Common examples are cardiogenic and noncardiogenic pulmonary edema, aspiration pneumonia, or extensive pulmonary hemorrhage. These disorders are associated with an increased work of breathing.

Types of Acute Respiratory Failure

3. CNS Causes:

- A variety of pharmacologic and structural of CNS are characterized by depression of the neural drive to breathe. This may lead to acute or chronic hypoventilation and hypercapnia.
- Examples include tumors or vascular abnormalities involving the brain stem, tetanus, and polio, CVA and an overdose of a narcotic or sedative.

Types of Acute Respiratory Failure

4. Peripheral nervous system:

- Disorders of the peripheral nervous system, respiratory muscles, and chest wall lead to an inability to maintain a level of **minute ventilation** appropriate for the rate of carbon dioxide production.
- Examples include Guillain-Barre syndrome, muscular dystrophy, and myasthenia gravis.

Clinical Manifestation of Respiratory Failure

- **Respiratory system:** difficult of breathing, dyspnea, tachypnea, orthopnea, hyperventilation, use of accessory muscle and nasal flaring.
- **Neurologic:** confusion, drowsiness, disorientation, and coma.
- **Cardiovascular:** tachycardia, cyanosis, diaphoresis, chest pain, peripheral vasodilatation with hypotension.

Diagnostic Evaluation of Respiratory Failure

- ABGs show hypoxemia, acidosis, alkalosis, and may hypercapnia.
- Chest x-ray shows pulmonary infiltration and atelectasis.
- Hematology may reveal increased WBCs, ESR and anemia (associated with hypoxia).
- Sputum study identifies organism.

Management of Respiratory Failure

Respiratory failure is a common and a life-threatening condition that demands prompt diagnosis and assessment and appropriate management.

- Correction of PaO_2 and $PaCO_2$
- Appropriate management of the underlying disease
- Bronchodilators: “Aminophylline, Aerosol”

Management of Respiratory Failure

- Analgesic: morphine sulfate.
- Diuretics: "Lasix" if overload is the cause.
- Steroids: "hydrocortisone- Solu-medrol"
- Histamine blockers: "Zantac, famotidine

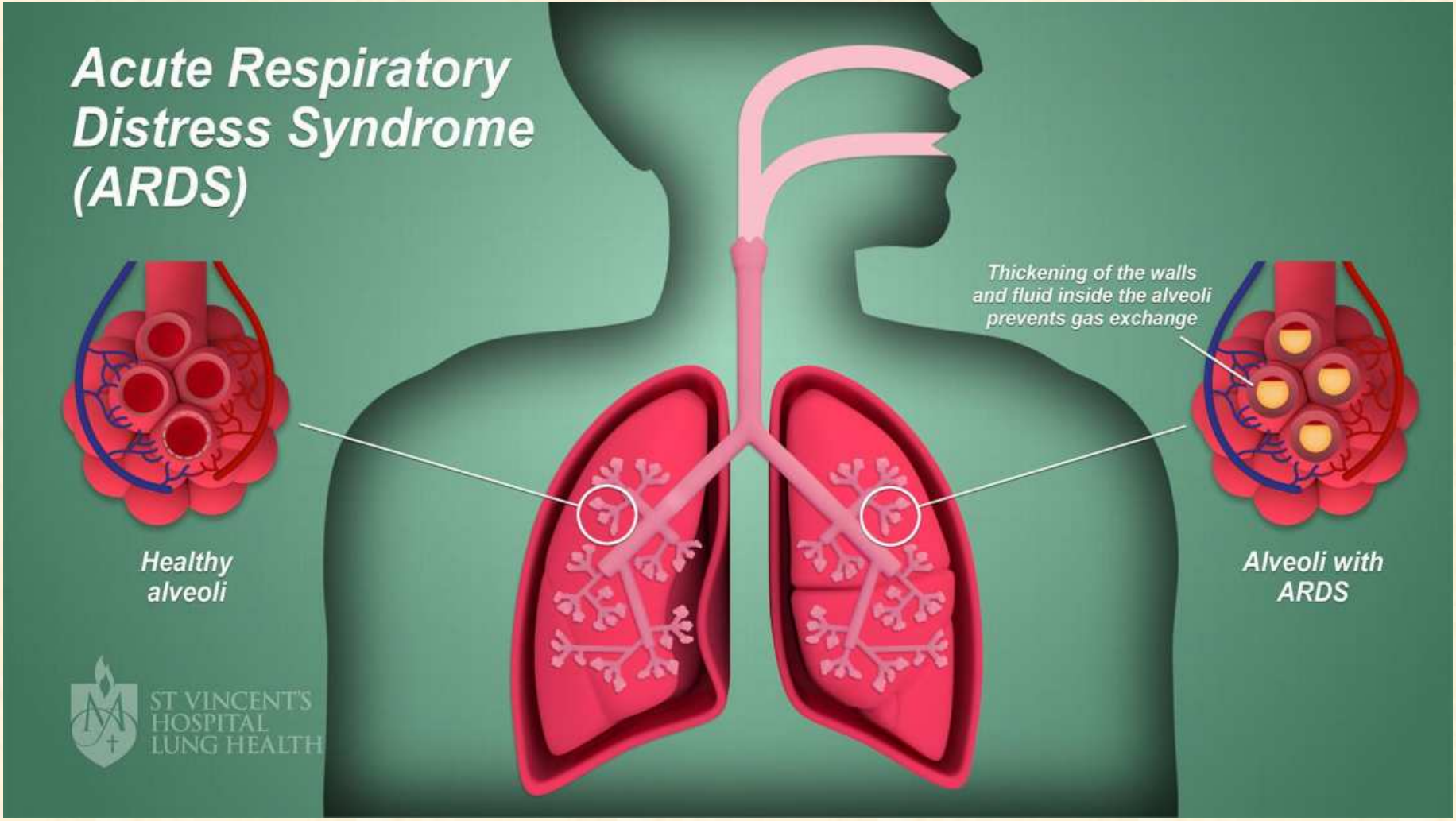
Nursing Interventions

- Maintain bed rest to reduce O2 requirement.
- Keep the patient in semi-fowlers position to promote chest expansion and ventilation.
- Administer O2 to reduce hypoxemia and relieve respiratory distress.
- Assess respiratory status to detect early signs of hypoxemia.
- Monitor and record vital signs, tachycardia and tachypnea may indicate hypoxemia.

Nursing Interventions

- Monitor pulse oximetry to detect a drop in SaO2.
- Provide suctioning, assist with coughing, and deep breathing exercises, and perform chest physiotherapy and postural drainage to facilitate removal of secretion.
- Report deteriorating: ABGS (PaO2 and PaCo2).
- CBC and Chemistry to detect electrolyte imbalance which result from the use of diuretics.

ACUTE RESPIRATORY DISTRESS SYNDROME (ARDS)

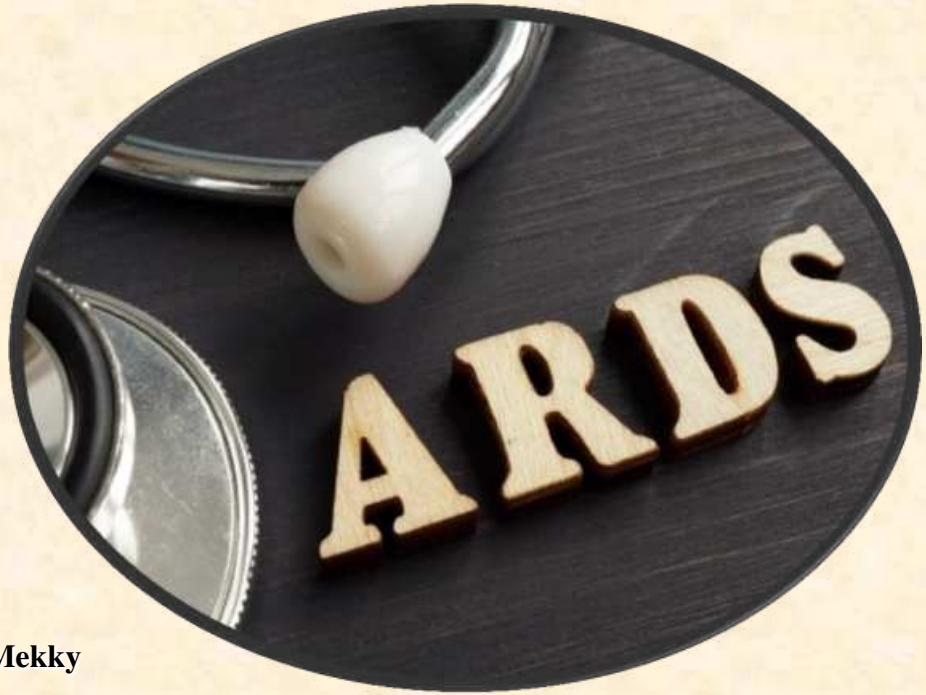


Acute Respiratory Distress Syndrome (ARDS)

- Acute respiratory distress syndrome (ARDS); previously known as Respiratory Distress Syndrome (RDS), adult respiratory distress syndrome, noncardiogenic pulmonary edema or shock lung, is an emergency medical condition occurring in critically ill patients characterized by widespread inflammation in the lungs.

Acute Respiratory Distress Syndrome (ARDS)

- The major cause of death in ARDS is non-pulmonary multiple-system organ failure, often with sepsis.
- ARDS has been associated with a mortality rate as high as 50% to 60%.



Acute Respiratory Distress Syndrome (ARDS)

Signs & Symptoms of ARDS:

- May include shortness of breath, fast breathing, reduced lung compliance, and hypoxemia. A chest x-ray frequently demonstrates generalized infiltrates in both lungs, which represent fluid accumulation in the lungs.
- These signs occur in the absence of left-sided heart failure.
- Chest auscultation reveals abnormal breath sounds; such as crackles that suggest fluid in the lungs.

Acute Respiratory Distress Syndrome (ARDS)

Signs & Symptoms of ARDS:

- Often the blood pressure is low as a result of hypovolemia secondary to leakage of fluid into the interstitial spaces and depressed cardiac output.
- Cyanosis (blue skin, lips, and nails) is frequently seen.





Acute Respiratory Distress Syndrome (ARDS)

Causes of ARDS:

- A wide range of causes are associated with development of ARDS, including:
  - Severe Pneumonia
  - Septic shock
  - Near Drowning
  - Chest Trauma
  - Aspiration of vomit, or chemical inhalation (Aspiration Pneumonia).

Acute Respiratory Distress Syndrome (ARDS)

Diagnostic Assessment of ARDS:

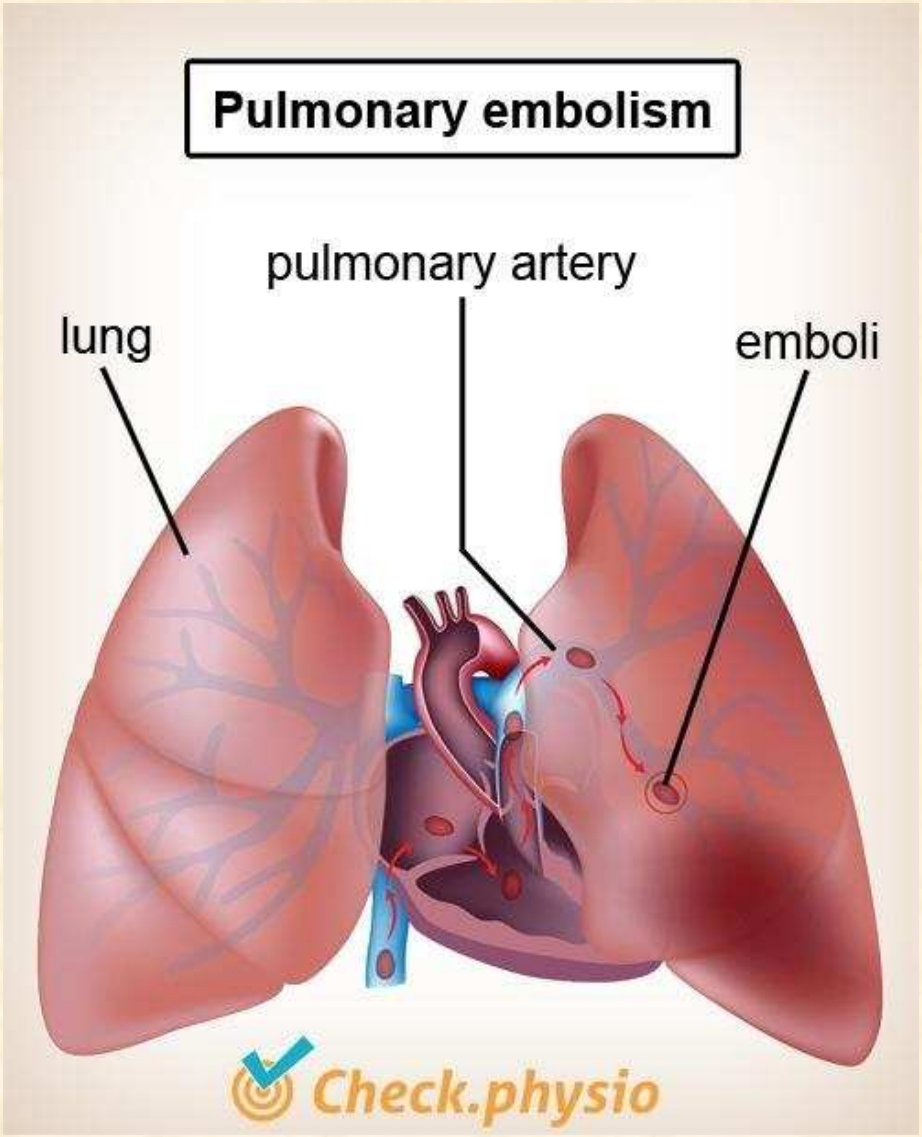
- ABGs show respiratory or metabolic acidosis and hypoxemia.
- Chest x-ray
- Blood and sputum cultures

Acute Respiratory Distress Syndrome (ARDS)

Treatment of ARDS:

1. Treatment of the underlying condition that caused ARDS is essential.
2. Mechanical ventilation with higher-than normal airway pressure, and PEEP to increase functional residual capacity. Avoid raising FiO_2 to reduce risk of O_2 toxicity.
3. Provide suction to remove of secretions.

Pulmonary Embolism (PE)



Pulmonary Embolism (PE)

- **PE** refers to the obstruction of the pulmonary artery or one of its branches by a thrombus that originate somewhere in the venous system or in the right side of the heart.
- **PE** is due to a blood clot or thrombus; but there are other types of emboli: air, fat, amniotic fluid, and septic (from bacterial invasion of the thrombus).
- Massive **PE** causing hemodynamic instability (marked decreased oxygen saturation, tachycardia and/or hypotension), and **death in 1 hour**.

Pulmonary Embolism (PE)

Risk Factors of PE:

- Deep venous thrombosis (DVTs) or pelvic vein thrombosis.
- Alterations in blood flow: immobilization (after surgery, injury or long distance air travel), pregnancy, and obesity.
- Factors affecting the properties of the blood (coagulant state).
- Estrogen-containing hormonal contraception.
- Genetic and acquired thrombophilia (means your blood can form clots too easily).

Pulmonary Embolism (PE)

Clinical Manifestations of PE: The symptoms of **PE** depend on the size of the thrombus and the area of the pulmonary artery occluded as the next:

- Dyspnea & Tachypnea are the most frequent symptom.
- Chest pain is common and is usually sudden.
- Circulatory instability as hypotension.
- Sudden death.

Pulmonary Embolism (PE)

Diagnostic Test Results of PE: Death from **PE** commonly occurs within 1 hour of symptoms; thus, early recognition and diagnosis are priorities.

- ABGs levels show respiratory alkalosis and hypoxemia.
- Pulmonary angiography shows location of embolism and filling defect of pulmonary artery.
- Physical findings: clinical signs and symptoms.
- Chest X-ray

Pulmonary Embolism (PE)

Treatment of PE:

- Bed rest with active and passive range of motion.
- Keep the patient in fowler position to enhance ventilation.
- Assist with coughing, and deep breathing exercises to mobilize secretions and clear airway.
- Administer O2 to enhance oxygenation.

Pulmonary Embolism (PE)

Treatment of PE:

Anticoagulant medication:

✓ Heparin (IV): stop further thrombus formation.

✓ Warfarin sodium (Coumadin): may be given simultaneously at the beginning or after 5-6 days of heparin therapy.

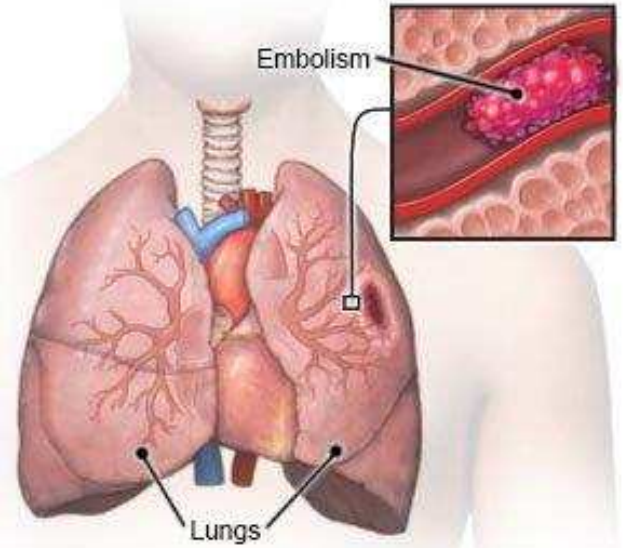
Thrombolytic agents:

✓ Streptokinase: to lyses thrombi; usually followed with heparin and warfarin treatment to prevent additional thrombus formation.

Pulmonary Embolism (PE)

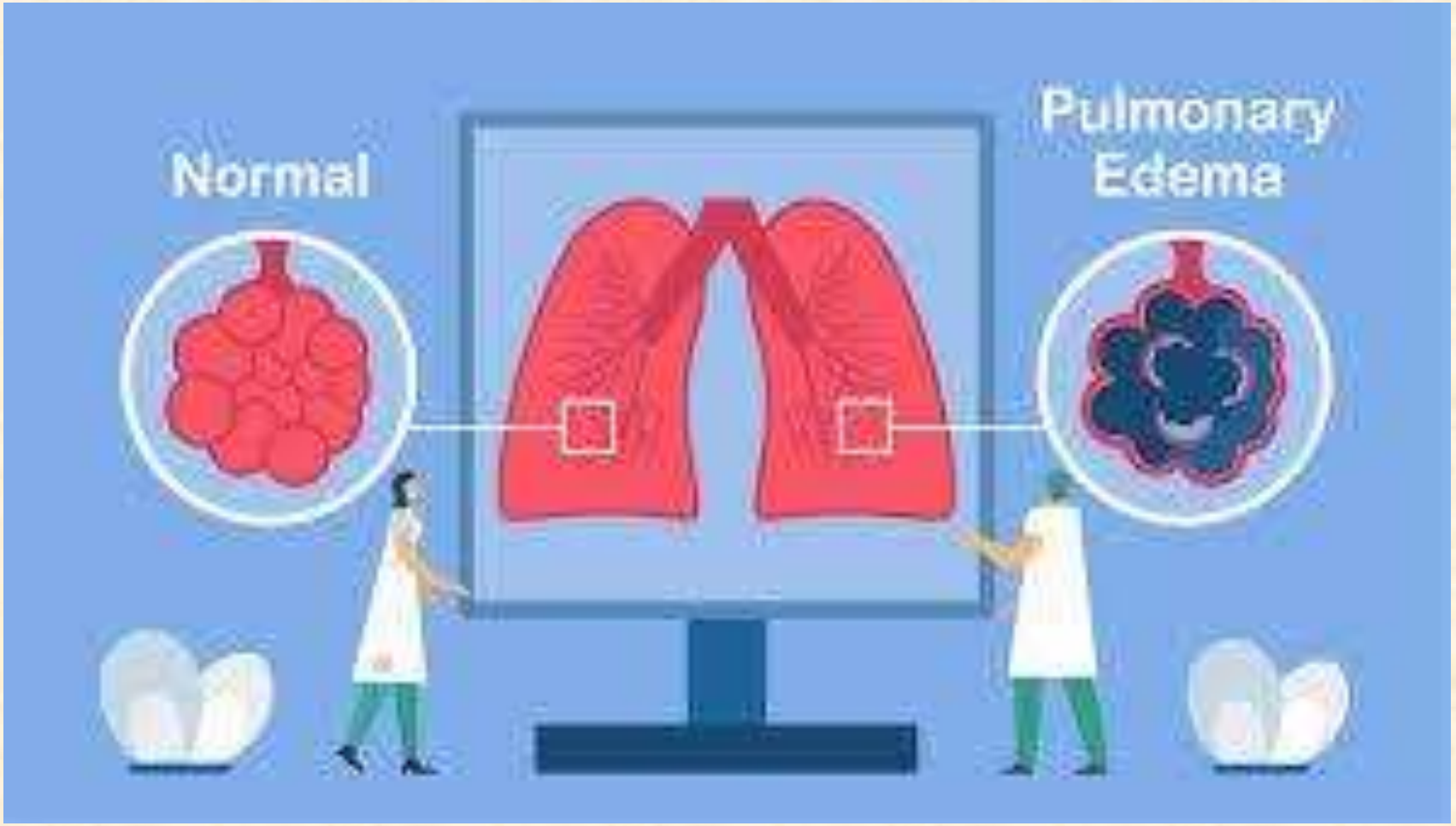
Treatment of PE:

- **Morphine:** given to reduce chest pain and anxiety.
- **Diuretics:** Lasix if right ventricular failure develops.
- **Surgical intervention:** embolectomy, inferior vena cava interruption.



Pulmonary Embolism

Pulmonary Edema



Pulmonary Edema

- Pulmonary Edema (some times called lung water) is the abnormal accumulation of fluid in the lungs. The fluid may accumulate in the interstitial spaces or in the alveoli.
- Pulmonary Edema is a life-threatening condition. It is often curable with urgent treatment and subsequent control of the underlying disorder.

Pulmonary Edema

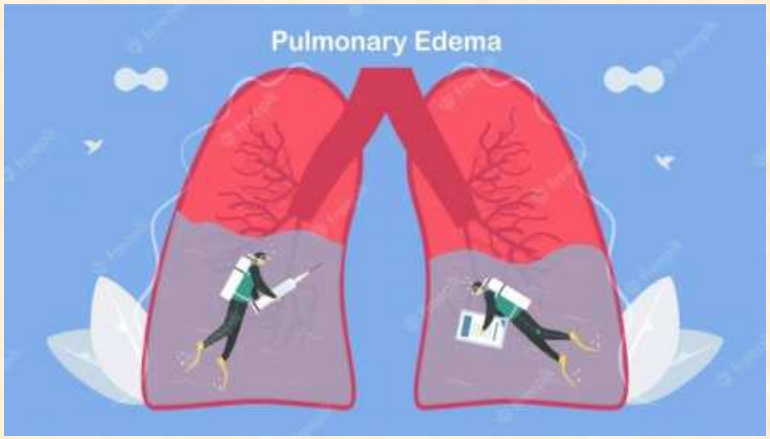
Causes of Pulmonary Edema:

- Pulmonary edema is a complication of a myocardial infarction, mitral or aortic valve disease, cardiomyopathy, or other disorders characterized by cardiac dysfunction.
- Also may caused as a result of noncardiac disorders, such as renal failure, liver failure, and Oncologic conditions that cause the body to retain fluid.

Pulmonary Edema

Clinical Manifestations of Pulmonary Edema:

- Extreme shortness of breath, severe difficult breathing.
- Grunting wheezing sounds with breathing and Nasal flaring.
- Anxiety and Excessive sweating.
- The nail beds become cyanotic.
- Cough, some times with blood.

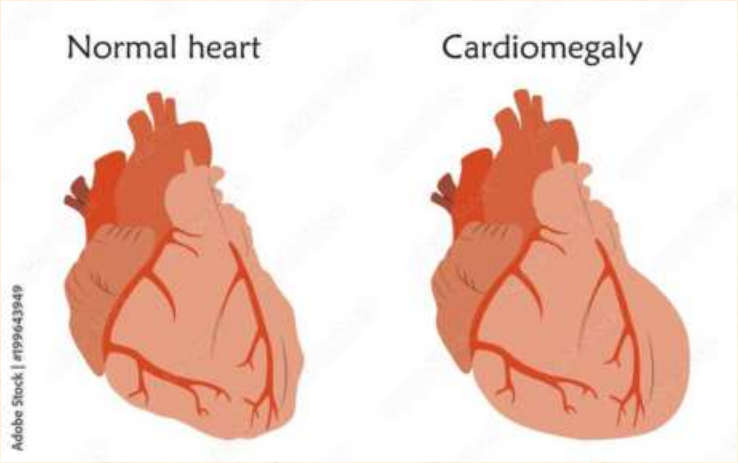


- The pulse is weak and rapid, and the neck veins are distended.
- The patient's oxygen levels (saturation) are significantly decreased.

Pulmonary Edema

Diagnostic Tests of Pulmonary Edema:

- ABG show low O_2 saturation.
- A chest X-ray may reveal fluid in or around the lung space, and enlarged heart.
- Ultrasound reveal Macrocardia (cardiomegaly).



Pulmonary Edema

Treatment of Pulmonary Edema:

- This is a medical emergency. Hospitalization and immediate treatment are required.
- Oxygen is given, by a mask or through endotracheal tube using mechanical ventilation (The use of PEEP is effective in reducing venous return).
- **Medications include diuretics such as:**

Pulmonary Edema

Treatment of Pulmonary Edema:

- ✓ Furosemide to remove fluid.
- ✓ Vasodilators to help the heart pump better.
- ✓ Morphine is administered intravenously in small doses (2 to 5 mg) to reduce peripheral resistance and venous return.
- ✓ Other medications to treat the underlying cardiac disorder.

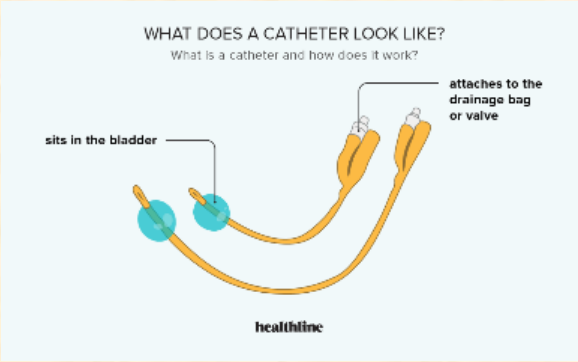
Pulmonary Edema

Treatment of Pulmonary Edema:

- Medical professionals use **DAMP MOIST Acronym** to help remember the initial treatment for pulmonary oedema:
- D: diuretics,
- A: aminophylline,
- M: morphine,
- P: position (upright position),

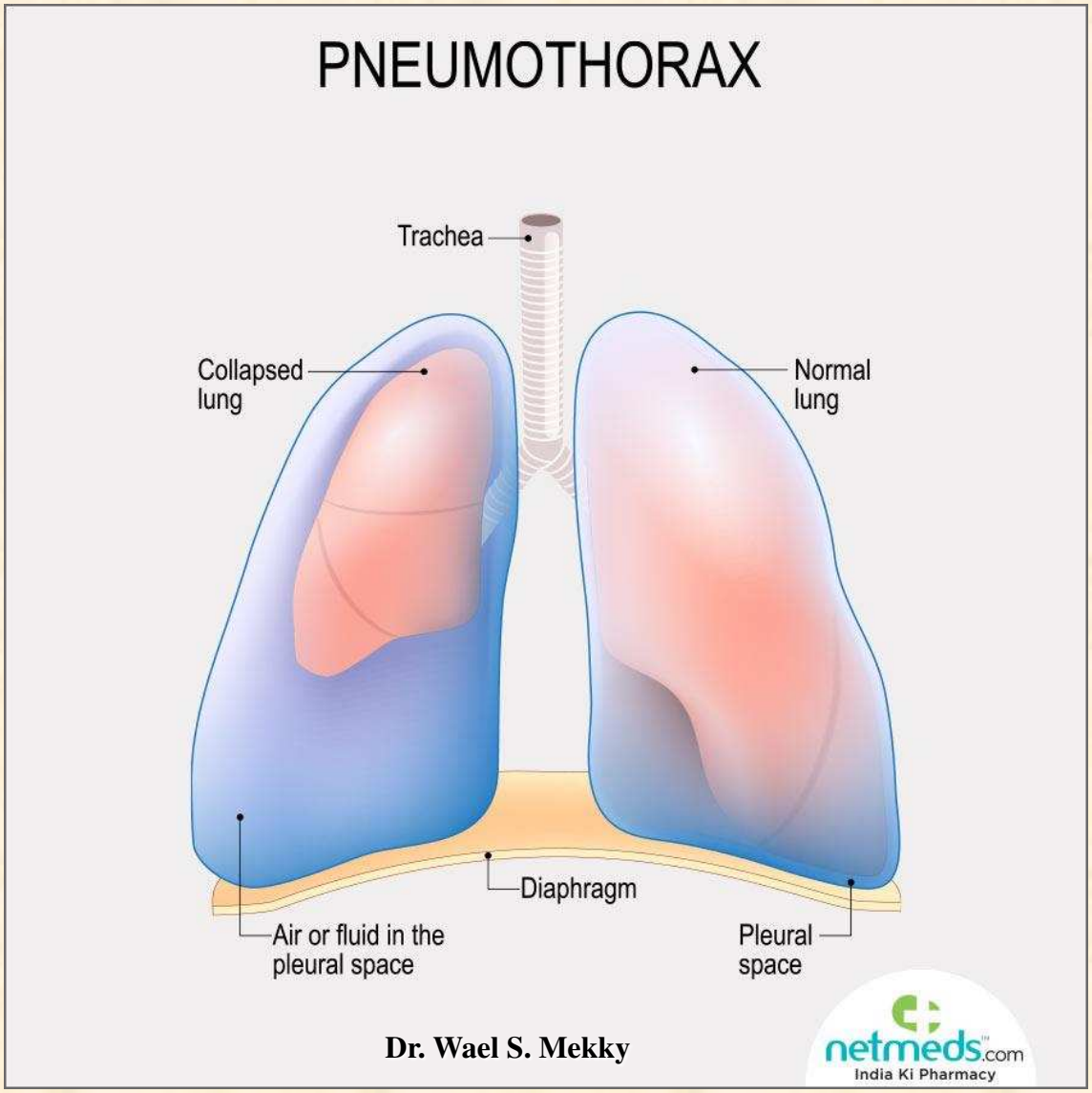
Pulmonary Edema

- M: monitoring blood pressure,
- O: oxygen,
- I: indwelling catheter (Foley catheter), and
- S: saturation
- T: tranquilizers (used to treat anxiety & reduces stress and tension).



❖ It is important to understand that the acronym represents the steps in treatment, but not necessarily the order in which they are administered.

Pneumothorax



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Pneumothorax

- Pneumothorax is defined as the presence of **air or gas** in the pleural cavity that occurs when the parietal or visceral pleura is penetrated and the pleural space is exposed to positive atmospheric pressure.
- Normally the pressure in the pleural space is negative; this negative pressure is required to maintain lung inflation.
- When either pleura is penetrated, air enters the pleural space, and the lung or a portion of it collapses.



Pneumothorax

Types of Pneumothorax:

1. Simple :

- Clinical condition stable, so can wait for CXR to confirm diagnosis

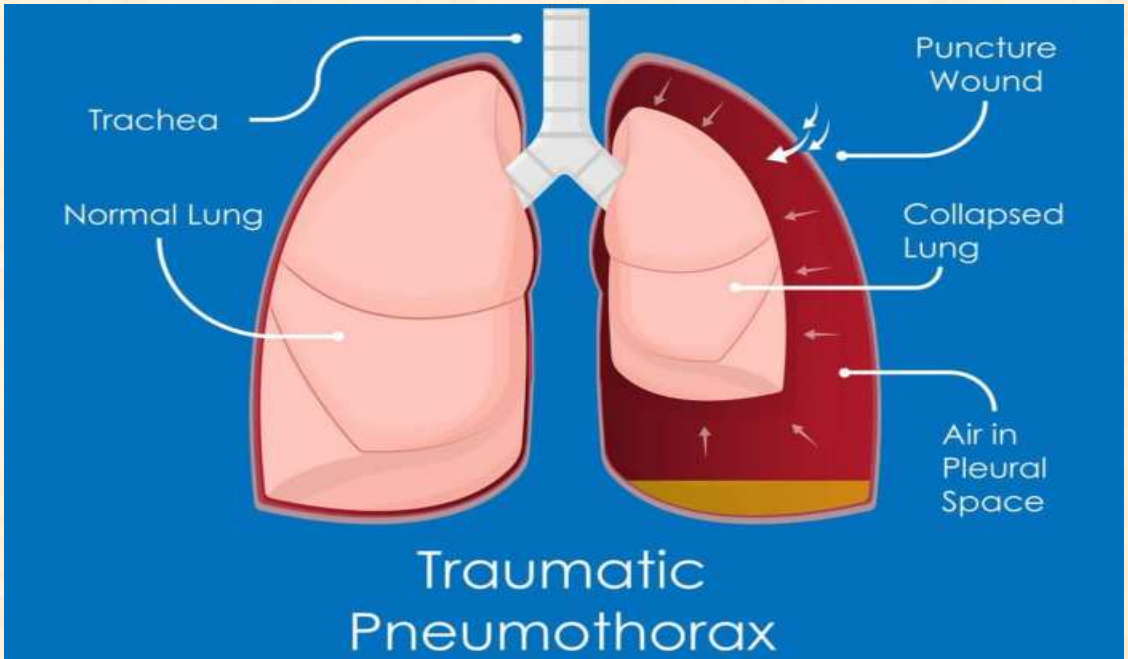
2. Tension :

- Progressive build up of air in the pleural space, causing a shift of the heart.
- Clinical condition unstable, so do not wait for CXR to confirm diagnosis.

Pneumothorax

Types of Pneumothorax:

3. Traumatic : Traumatic pneumothorax occurs when air escapes from a laceration in the lung itself and enters the pleural space through a wound in the chest wall. It can occur with blunt trauma (eg, rib fractures) or penetrating chest trauma.



Pneumothorax

Causes of Pneumothorax:

- **Spontaneous**
- **Traumatic**

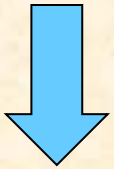
✓ With rib fractures

✓ Penetrating chest trauma

- **Pre-existing lung abnormality**

✓ Pulmonary fibrosis ✓ Asthma ✓ Vasculitis ✓ Pulmonary metastases close to edge of lung.

Vasculitis: is the name of a group of conditions that cause inflammation of the blood vessels.



Pneumothorax

Medical Management of Pneumothorax:

- Medical management of pneumothorax depends on its cause and severity. **The goal of treatment** is to evacuate the air or blood from the pleural space.
- A small chest tube (28 French) is inserted near the second intercostal space; this space is used because it is the thinnest part of the chest wall, minimizes the danger of contacting the thoracic nerve, and leaves a less visible scar.

Pneumothorax

Medical Management of Pneumothorax:

- If the patient also has a hemothorax, a large-diameter chest tube (32 French or greater) is inserted, usually in the fourth or fifth intercostal space at the midaxillary line.
- If a tension pneumothorax is suspected, the patient should immediately be given a high concentration of supplemental oxygen to treat the hypoxemia, and decompressed by inserting a large-bore needle (14-gauge) at the second intercostal space, midclavicular line on the affected side.



ANY QUESTIONS/ COMMENTS???

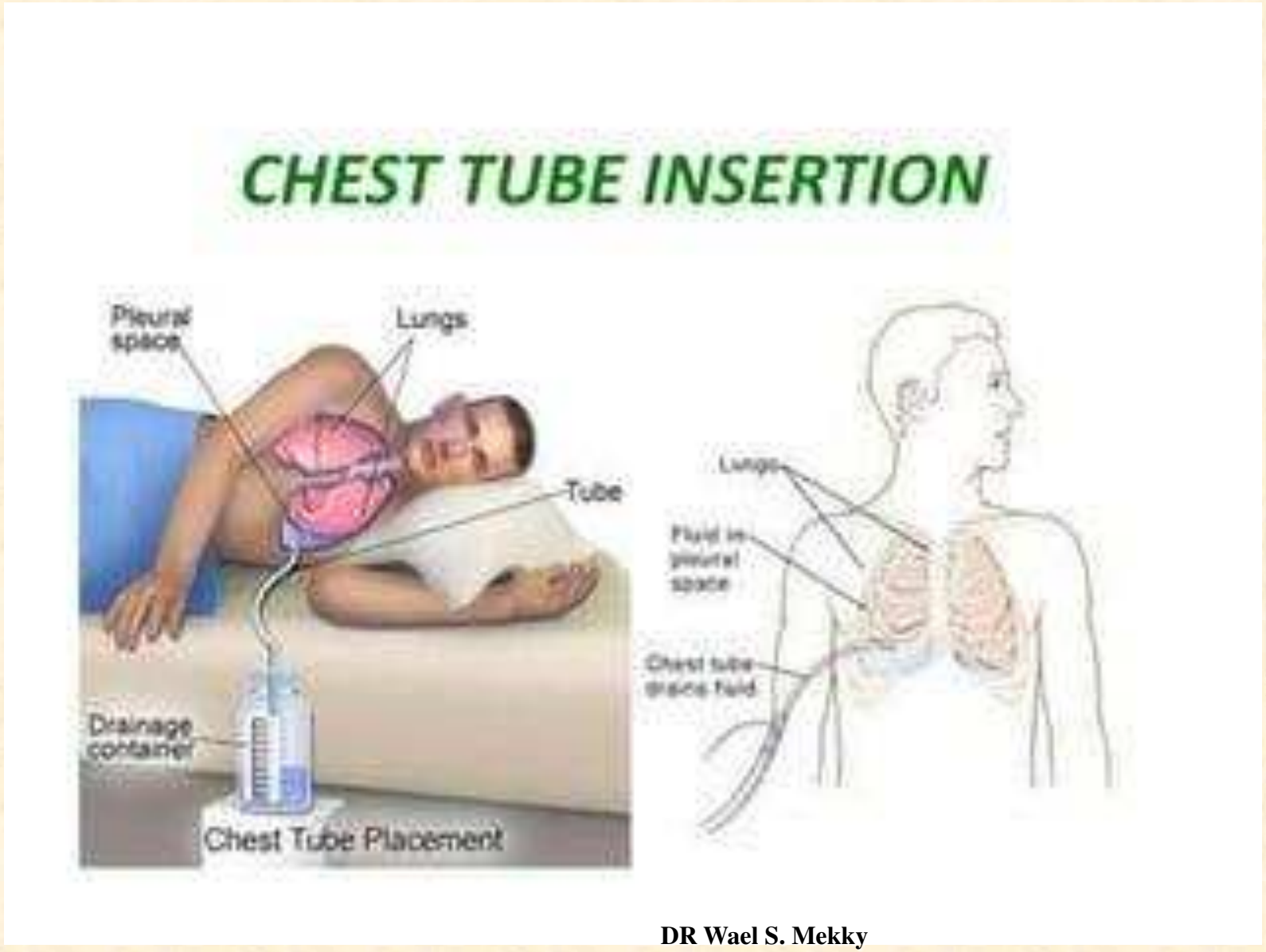
NURSING MANAGMENT OF CHEST TUBES

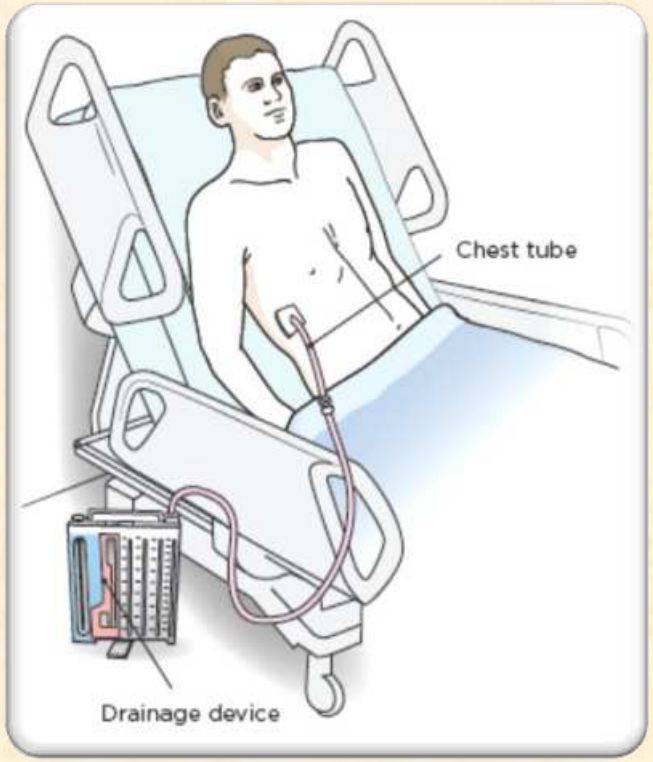


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Chest Tube


CHEST TUBE INSERTION
Pleural space
Lungs
Tube
Drainage container
Chest Tube Placement
Lungs
Fluid in pleural space
Chest tube drains fluid


Chest tube
Drainage device





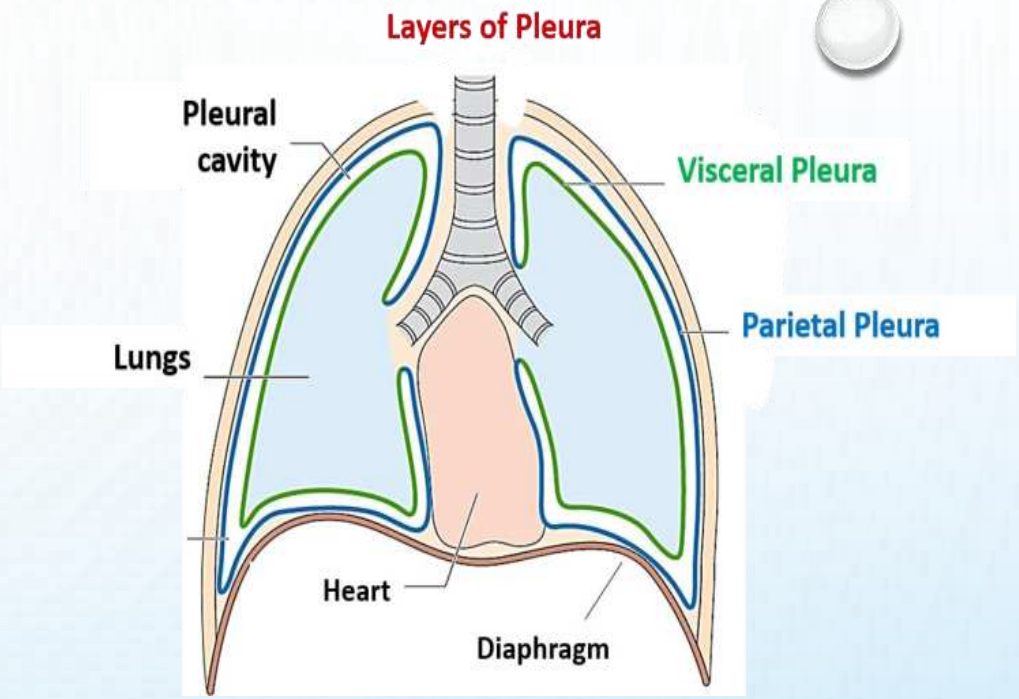






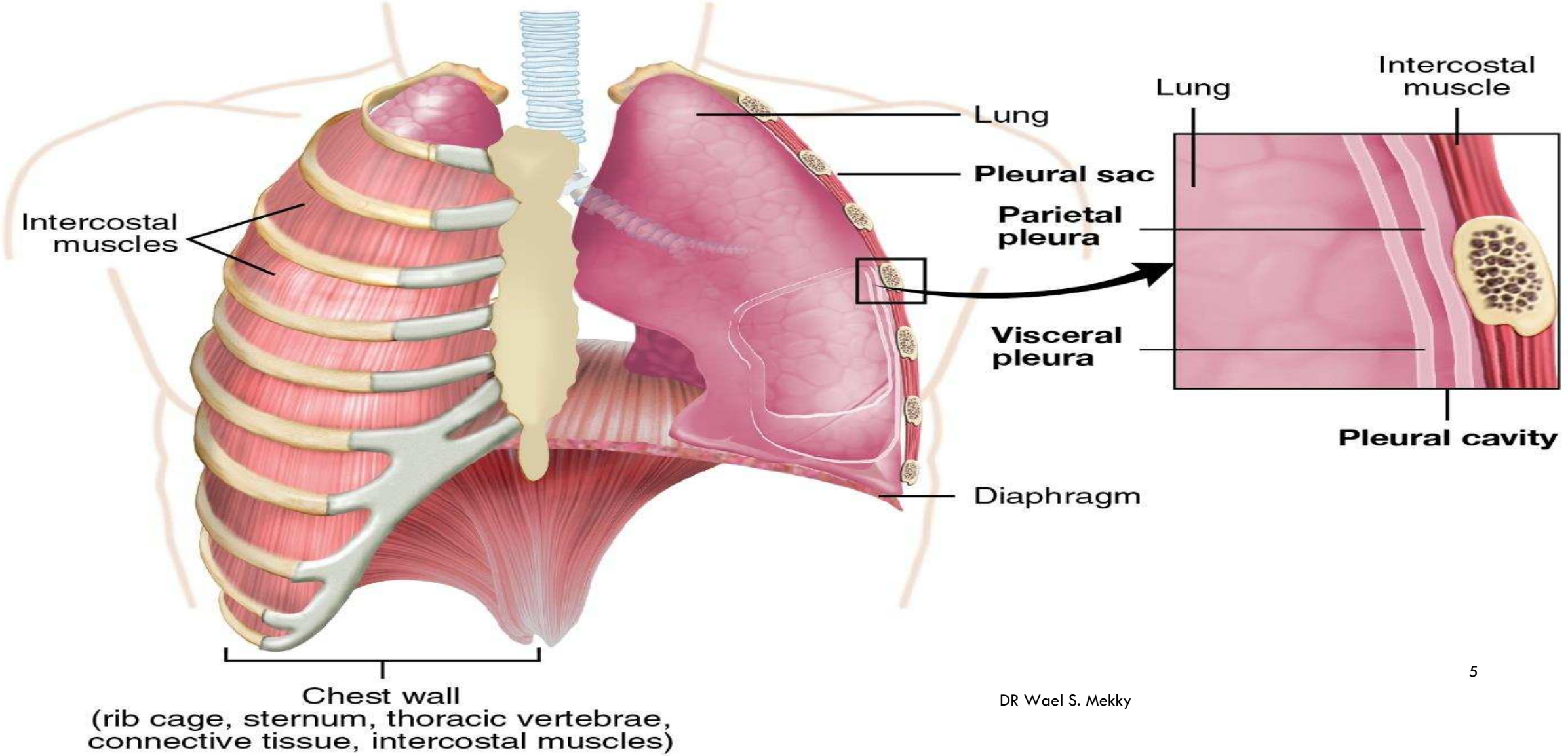
DR Wael S. Mekky

LUNG ANATOMY

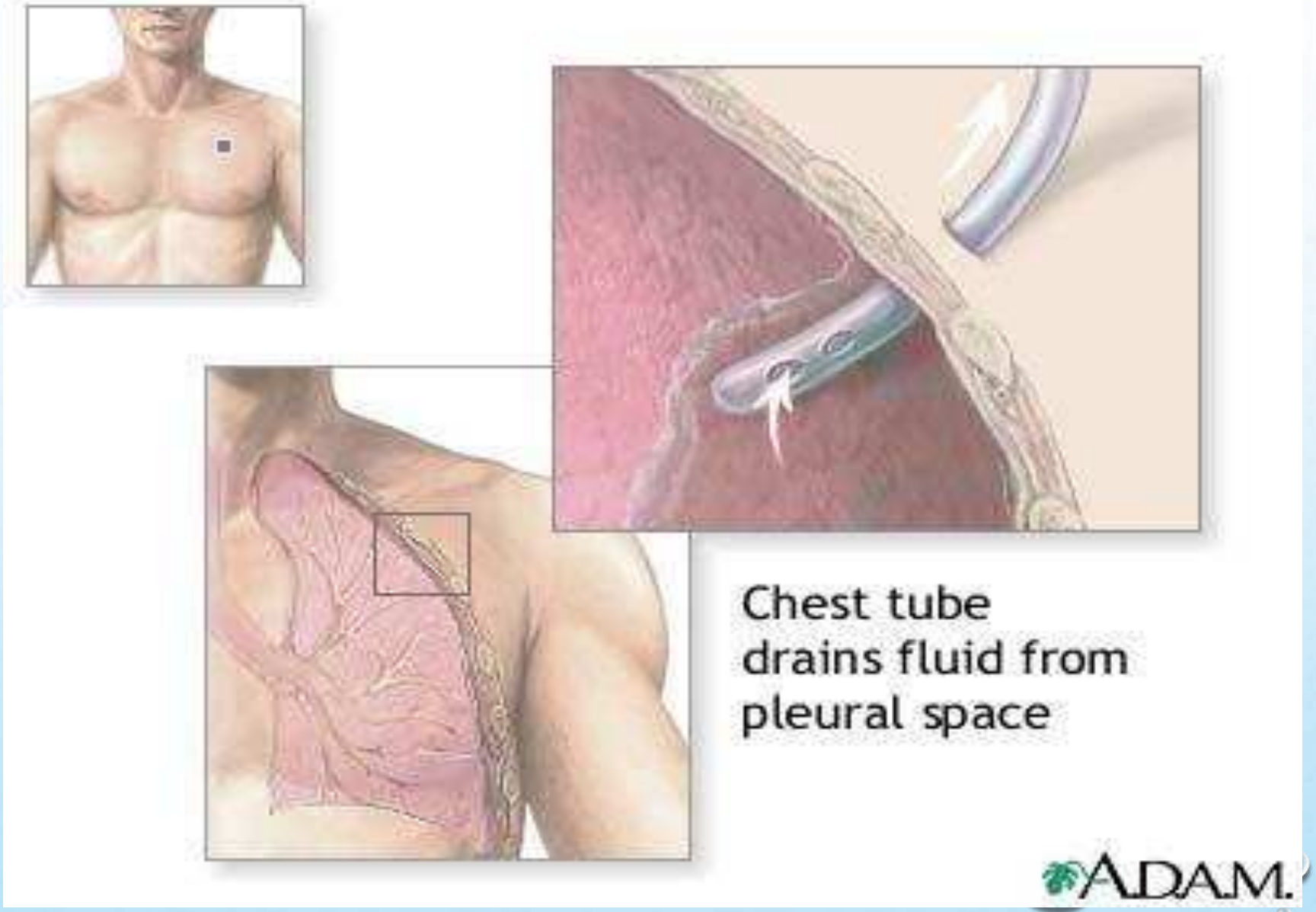


- **Parietal pleura**: Inside membrane of the rib cage
- **Pulmonary or visceral pleura**: Outside membrane of the lungs
- The two membranes are separated by a lubricating fluid (**pleural fluid)** which enables them to slide against each other. Lung can expand and contract
- Vacuum or “ negative pressure ” in the **pleural space** keeps the two pleurae together. If fluid or air enters the pleural space, lungs cannot fully expand during each respiratory cycle.

LUNG ANATOMY



CHEST TUBE PLACEMENT



INDICATIONS FOR CHEST TUBE

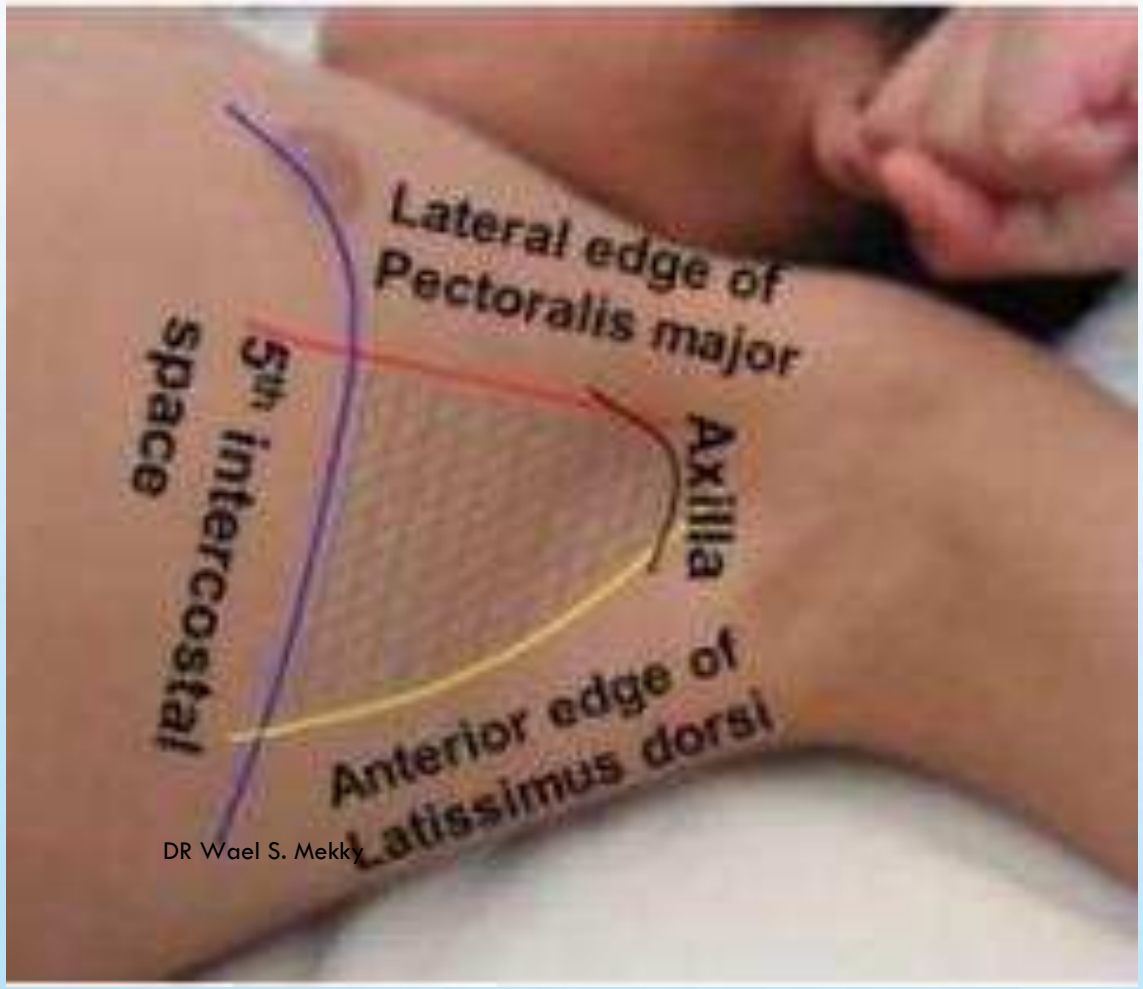
- **PNEUMOTHORAX**: Air In The Pleural Space (Trauma, Lung Disease)
- **HEMOTHORAX**: Blood In The Pleural Space (Blunt Or Penetrating Trauma)
- **HEMOPNEUMOTHORAX**: Air & Blood In The Pleural Space
- **TENSION PNEUMOTHORAX**: Occurs When Injured Tissue Forms A One Way Valve Or Flap Allowing Air To Enter But Preventing It From Escaping Naturally, (Emergent Situation)

INDICATIONS FOR CHEST TUBE

- ***PLEURAL EFFUSION***: Excessive Fluid In Pleural Space
- ***EMPYEMA***: Collection Of Pus Due to Infection
- ***CHYLOTHORAX***: Lymphatic Fluid Accumulation In The Pleural Space
- ***PREVENT POST OPERATIVE COMPLICATIONS***: Post Cardiac Surgery Or Chest Trauma.

INSERTION SITE

"Safe Triangle"



- Recommended by The British Thoracic Society guidelines.
- Minimizes risk of injury to the internal mammary artery, muscle or breast tissue.

All Experts agree:

Tube placed in the midaxillary line between **4th and 5th** Ribs should successfully drain fluid or air.

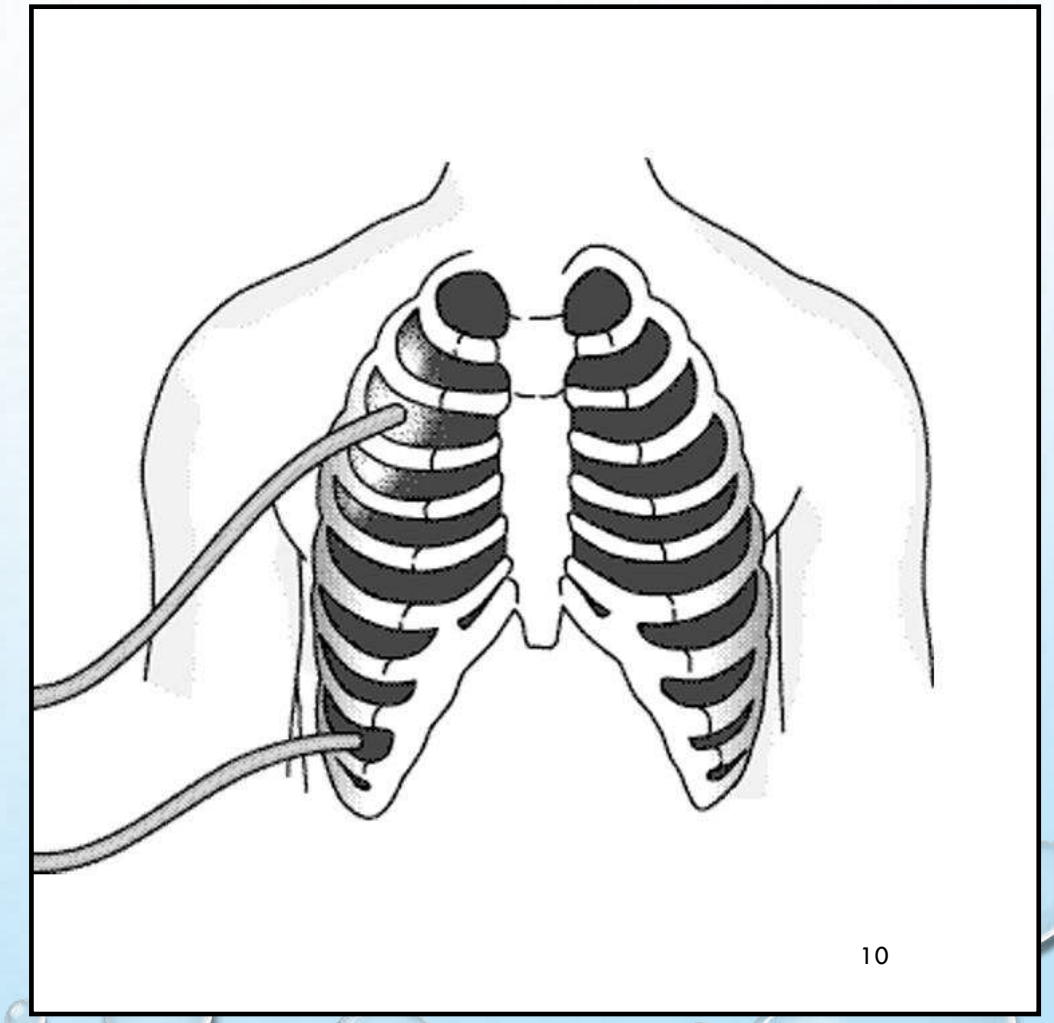
INSERTION SITE

Superior Tube Placement:

Removal of Air (ie. Pneumothorax)

Inferior Tube Placement:

Removal of Fluids (ie. Hemothorax, Empyema)

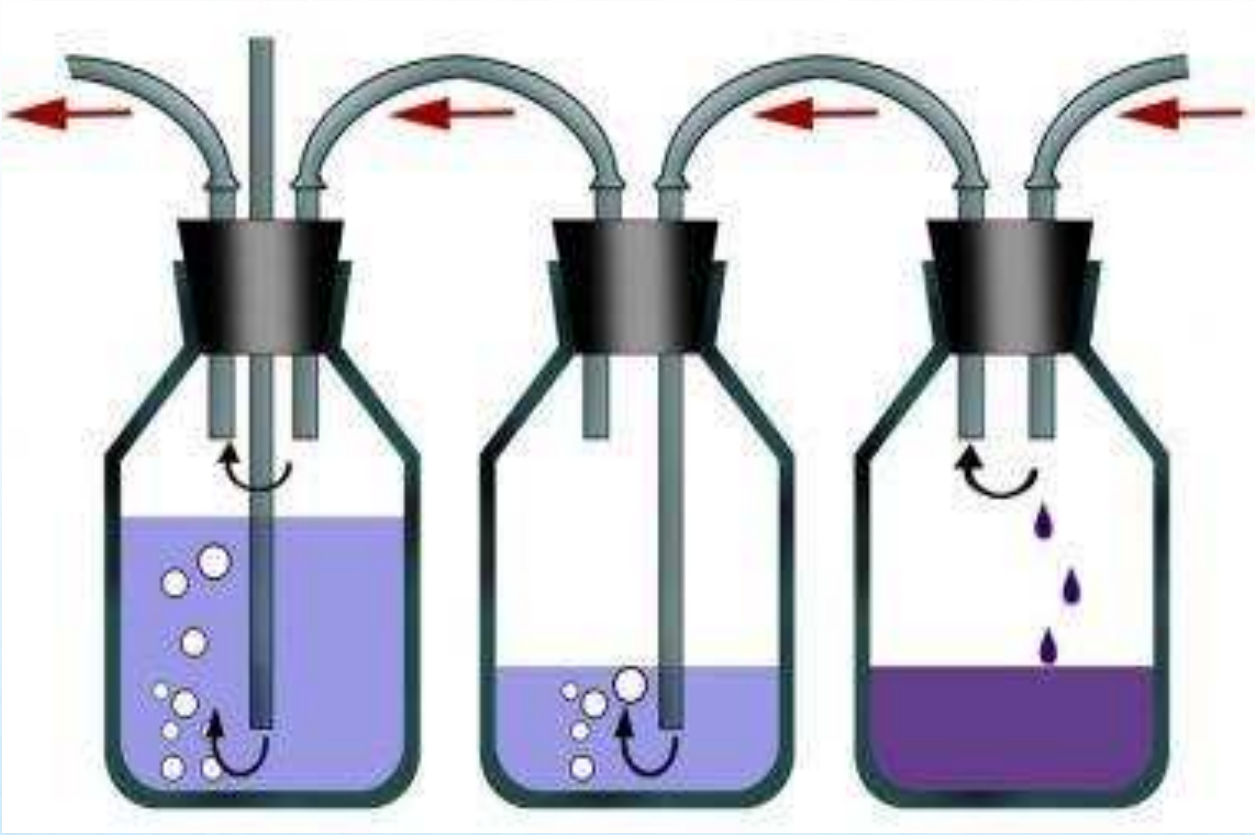


CHEST TUBE SYSTEM

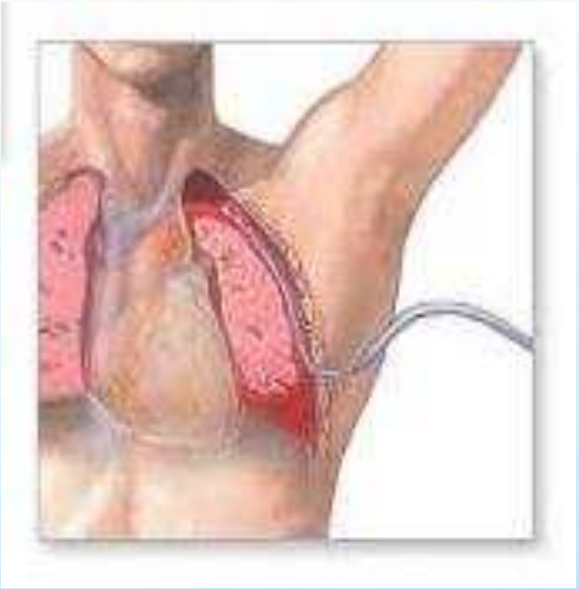
3 BOTTLE SYSTEM

Suction





Patient

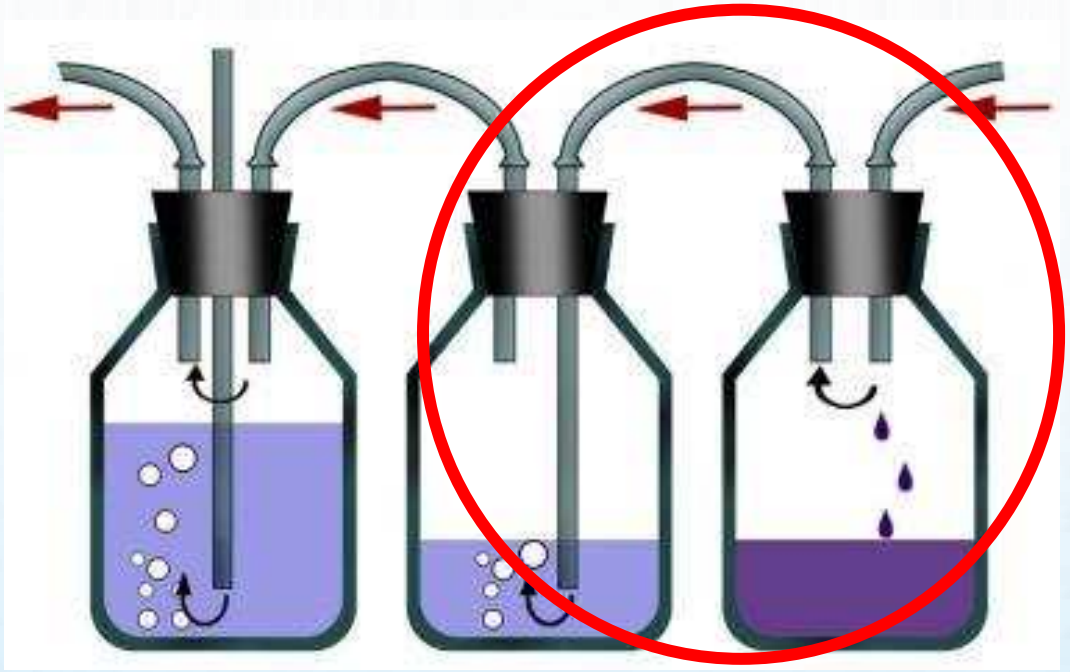


Suction Control | **Water Seal** | **Collection Bottle**

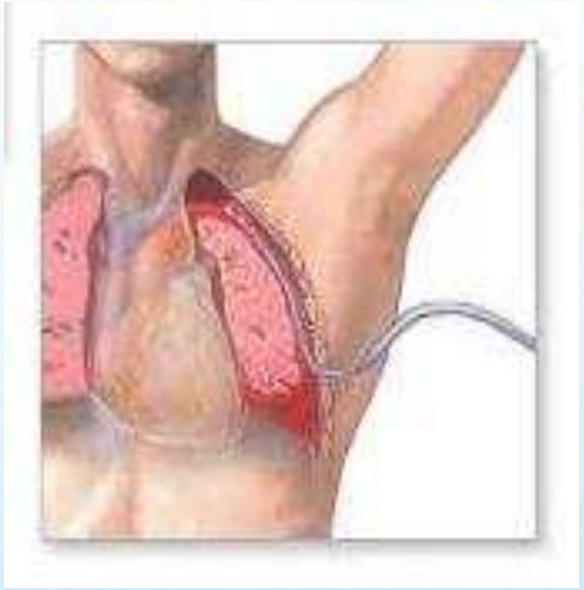
COLLECTION BOTTLE AND WATER SEAL

SUCTION





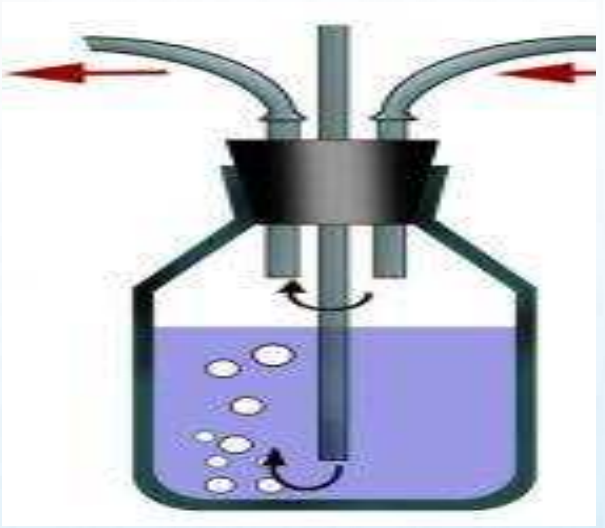
Patient



Collection Chamber: Collects drainage

Water Seal: Maintained at 2cm, One way valve; Prevents air or fluid from returning to the chest.

SUCTION CONTROL



- The **Depth Of The Water** In The Suction Bottle

  Determines The Amount Of Negative Pressure That Can

  Be Transmitted To The Chest, **NOT The Reading On The**

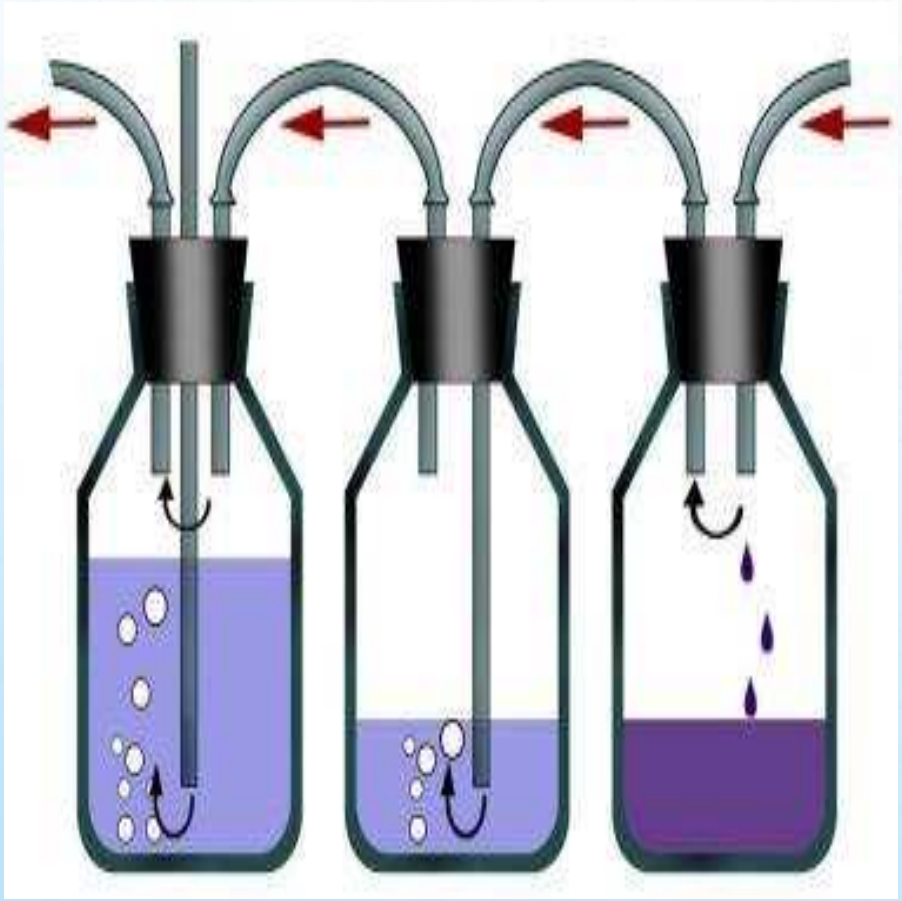
  Vacuum Regulator.



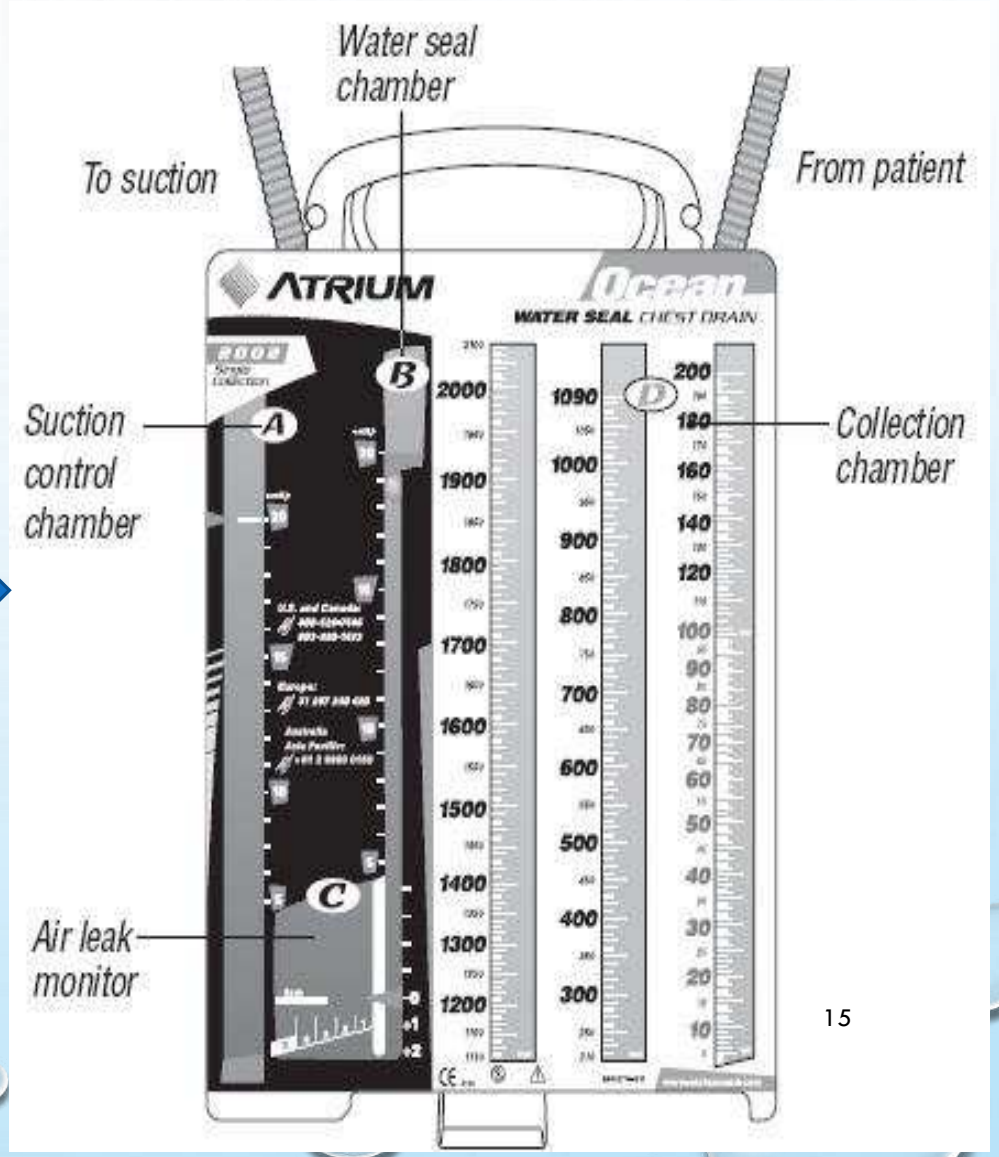
- Higher Negative Pressure Can Increase The Flow Rate

  Out Of The Chest, But It Can Also Damage Lung Tissue

CHEST TUBE DRAINAGE UNIT



Modern Day



ATRIUM DRAINAGE SYSTEM



A: Collection Chamber

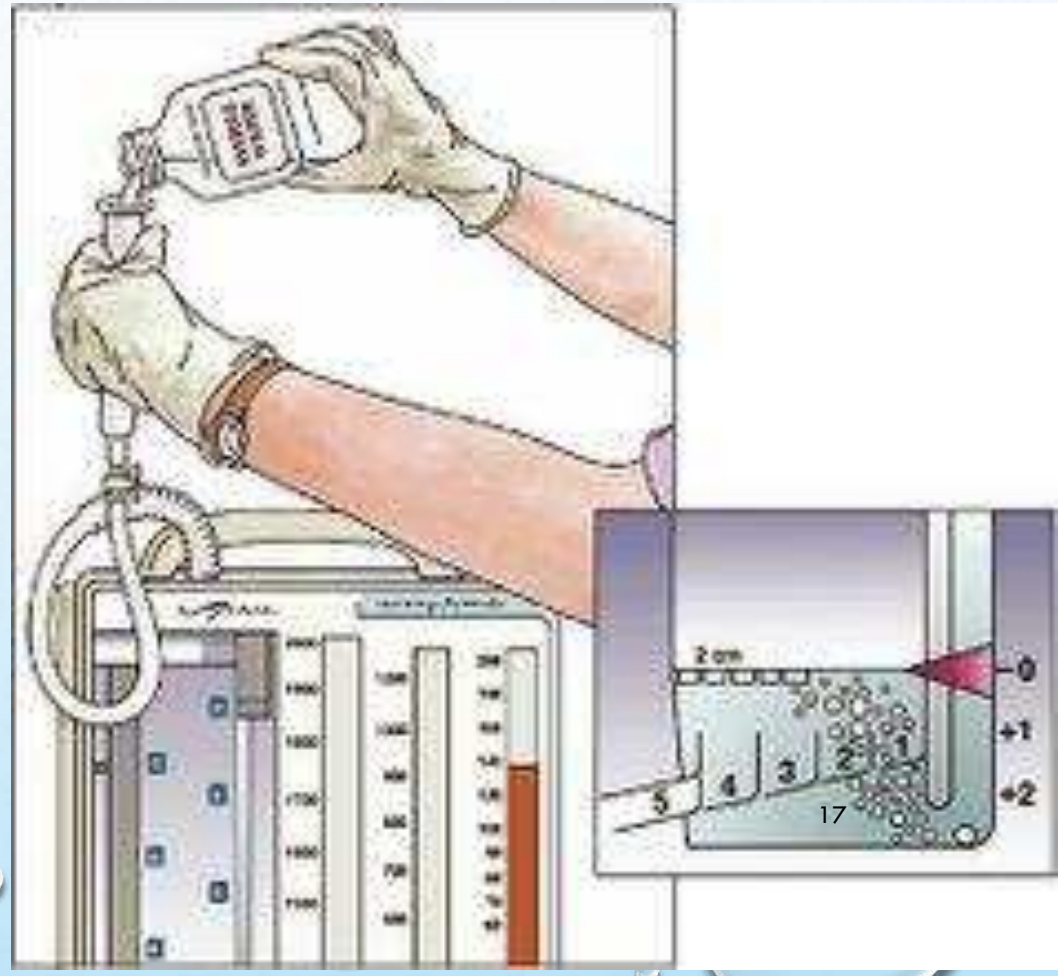
B: Water Seal

C: Suction Control

SETTING UP THE SYSTEM

1. Fill the water seal chamber with 2cm of sterile water, water will turn blue
2. Connect patient tube to thoracic catheter
3. Set Up Suction

   - **Water Suction**: Fill chamber to 20cm, connect the drain to vacuum, and slowly increase vacuum until ***gentle bubbling*** appears in the suction control chamber
   - **Dry suction**: Set the dial to prescribed pressure, connect to suction source and turn to **-80mmHg or higher if needed**.



NURSING ASSESSMENTS AND INTERVENTIONS

ASSESSMENT

USE THE 

ACRONYM

S = Site
T = Tubing
O = Output
P = Patency

STOP



S = Site

Check Site:

- Ensure tube is in place,
- No S&S of infection
- Dressing: change q48h & PRN wet or soiled. Sterile technique, occlusive dressing.
- ***Check for Subcutaneous emphysema***

Subcutaneous Emphysema (Air Underneath the Skin)



Subcutaneous emphysema (air trapped under skin)

a. Palpate

b. Continuous Monitoring

Check the Dressing & Site



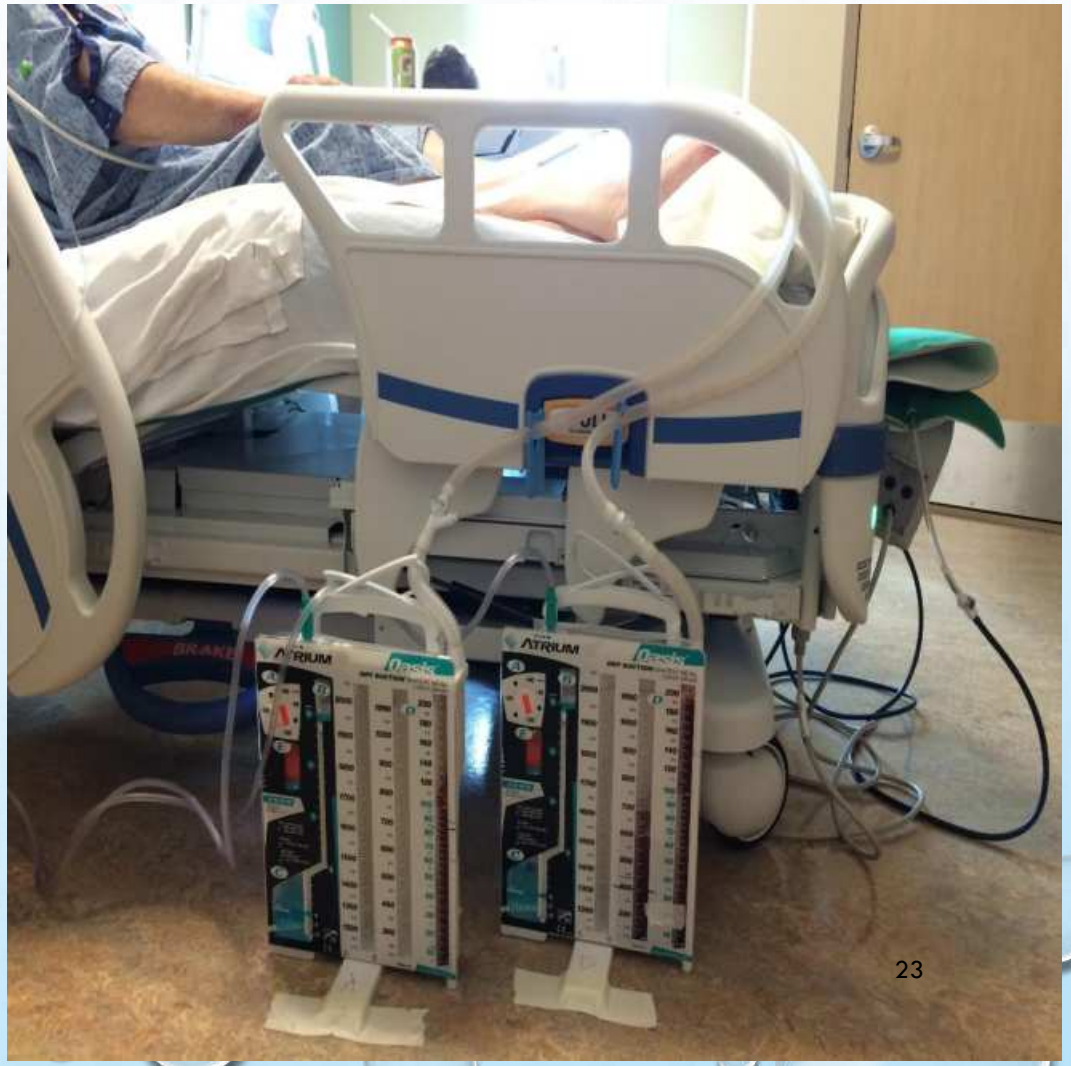


sTop

T = Tubing

- Connections are secured
- No kinks
- Keep the drainage system

below patient's level



stOp

O = Output

- Check the level regularly.
- Document amount, type, and color.



STOP

P = Patency (A: water seal)

- Assess the **water seal** with the **suction off**
- If water seal level is **too high**, it will be more difficult for air to leave the chest (**aspirate water** via grommet)
- If the water seal level is **too low**, it leaves the water seal chamber at risk for exposure to air and can cause a pneumothorax (**add sterile water** via grommet)

STOP

P = Patency (B: Bubbling)

Bubbling means there is a **leak in the system** unless the patient has a pneumothorax.

- **Intermittent bubbling** may indicate air leak from pleural space, will resolve when lung re-expands.
- **Continuous bubbling** suspect a leak in the system.

STOP

P = Patency (C: fluctuations)

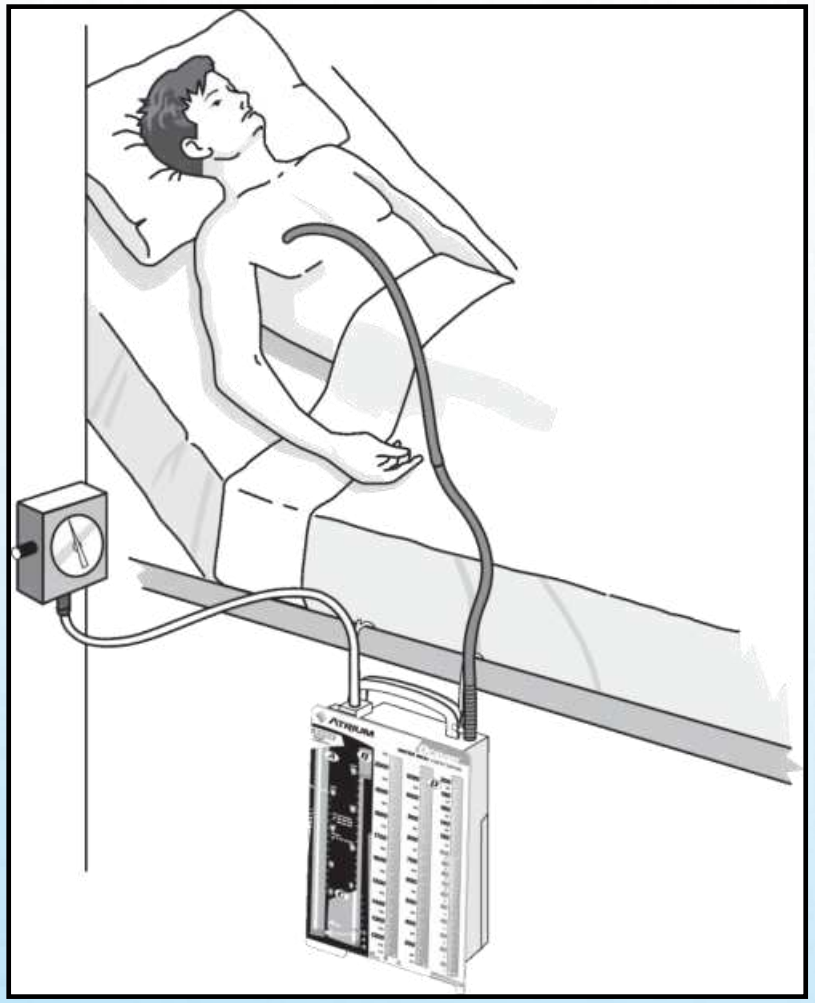
- Normal:
  - Inspiration the water seal level will reach **-10cm**
  - Expiration the water seal level will reach **-4cm** of negative pressure normally
- If **fluctuations stop**, the tubing may be **obstructed** (kinks, pt lying on tubing).

ASSESS PULMONARY STATUS

- Respiration Rate
- Work of Breathing
- Breath Sounds
- Continuous SpO_2 Monitoring



AT THE BEDSIDE



- Always keep drain **BELOW** the chest for gravity drainage

- Fluid moves from an area of higher pressure to an area of lower pressure.

ITEMS TO BE KEPT AT BEDSIDE

- 2 CLAMPS
- VASELINE GAUZE
- 4X4 GAUZE DRESSING
- WIDE TAPE
- NEW DRAINAGE SYSTEM

CLAMPING



Clamp Only When

- ✓ Changing the device (the drainage system)
- ✓ Checking for an air leak
- ✓ Tubing is disconnected
- ✓ Checking lung re-expansion prior to chest tube removal

DO NOT Clamp

- Ambulating Patient (will increase risk of tension pneumothorax)
- Transport – Disconnect suction tubing at Suction Control Stop cock, leave open.

CHEST TUBE REMOVAL

INDICATIONS:

- Improved respiratory status
- Symmetrical rise and fall of chest
- Bilateral Breath Sounds
- Decreased chest tube drainage
- Absence of bubbling in the water seal chamber during inspiration
- Improved CXR findings.

CHEST TUBE REMOVAL

Process:

- **Premedication given** to relieve pain.
- **Supplies:** sterile gloves, gown, mask, dressing supplies, sterile suture removal kit, wide occlusive tape.
- **Semi-Fowlers** position
- Clamps chest tube, and then chest tube removed
- Occlusive dressing applied
- CXR – ensure lung remains fully inflated.




THANK YOU

Acute Alteration in Hemodynamic Functions



BY: Dr. Wael S. Mekky

HEMODYNAMIC MONITORING

Introduction to Hemodynamic Monitoring

- Hemodynamics is concerned with the forces generated by the heart and the resulting motion of blood through the cardiovascular system.
- It is a term that used to describe the intravascular pressure and flow that occurs when the heart muscle contracts and pumps blood throughout the body.

Introduction to Hemodynamic Monitoring

- Hemodynamics is the forces involved in blood circulation. Hemodynamic monitoring started with the estimation of heart rate using the simple skill of 'finger on the pulse' and then moved on to more and more sophisticated techniques like stethoscope, sphygmomanometer, ECG etc.
- The status of critically ill patients can be assessed either from non-invasive single parameter indicators or various invasive techniques that provide multi-parameter hemodynamic measurements.

Introduction to Hemodynamic Monitoring

- Because the vascular system is a closed circuit; the pressure and flow variations in the venous compartment will necessarily affect the arterial compartment and vice versa.
- Therefore, critically ill patients require continuous assessment of their cardiovascular system to diagnose and manage their complex medical conditions. This is most commonly achieved by the use of direct pressure monitoring systems, often referred to as hemodynamic monitoring.

Introduction to Hemodynamic Monitoring

- Central Venous Pressure (CVP), Pulmonary Artery Pressure, and Arterial Pressure monitoring are common forms of invasive hemodynamic monitoring.

Definition of Hemodynamic Monitoring

- Hemodynamic monitoring is the intermittent or continuous observation of physiological parameters pertaining to the circulatory system with a view to early detection of need for therapeutic interventions.

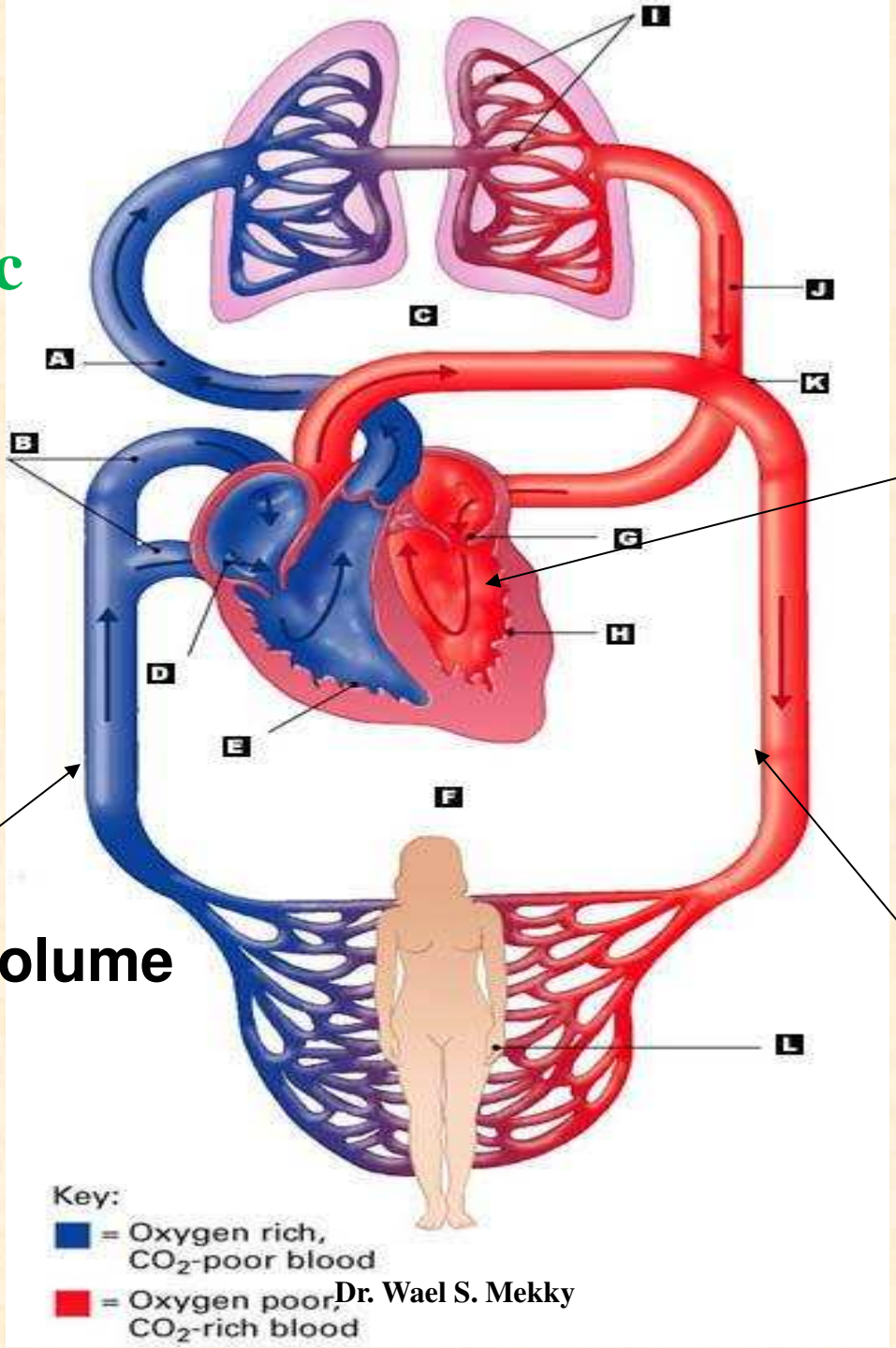
Definition of Hemodynamic Monitoring

- Hemodynamic monitoring refers to measurement of pressure, flow and oxygenation of blood within the cardiovascular system.

OR

- Using invasive technology to provide quantitative information about vascular capacity, blood volume, pump effectiveness and tissue perfusion.

4 factors that affecting the hemodynamic conditions



Myocardial contraction and heart rate

Intravascular volume

Vasoactivity

Factors Affecting Hemodynamic conditions

(1) Intravascular volume: the amount of fluid circulating in the vasculature. This can be affected by dehydration, diuresis, and volume overload due to heart or kidney failure.

(2) Vasoactivity: affected by hormone eg., Angiotenson II, Epinephrine, Norepinephrine, and Vasopressin.

(3) Myocardial contraction and (4) HR: affected by exercise, stress and pharmaceutical agents or cardiac disease.

Manifestations of Altered Hemodynamic

- Symptoms of weakness, pallor, confusion, cold clammy skin, diminished or absent pulses, cardiac arrhythmias, low arterial blood pressure, and decreased cardiac output.
- New murmurs and abnormal heart sounds.

Indications For Hemodynamic Monitoring

- Continuous, real-time blood pressure monitoring.
- Planned pharmacologic or mechanical cardiovascular manipulation.
- Repeated blood sampling.
- Determination of volume responsiveness from systolic pressure or pulse -pressure variation.
- Supplementary diagnostic information from the arterial waveform.

Classic Hemodynamic Monitoring Methods

- **Non-invasive Hemodynamic Monitoring Methods:**
  - Pulse Rate & Quality
  - Blood Pressure
  - Skin Temperature / Color
  - Capillary Refill
  - Pulse Oximeter
  - Urine Output (UOP)
- **Invasive Hemodynamic Monitoring Methods:**
  - Arterial Line (Catheter)
  - Central Veinous Catheter (CVC)
  - Pulmonary Artery Catheter (PAC)

Types of Invasive Hemodynamic Monitoring

Invasive Hemodynamic Monitoring:

1. Arterial pressure monitoring
2. Central venous pressure monitoring
3. Pulmonary artery pressure monitoring

(1) Arterial Pressure Monitoring

(1) Arterial Pressure Monitoring

- **Arterial pressure catheter:**

- Arterial line placement is a common procedure in various critical care settings.

- Intra-arterial blood pressure (BP) measurement is more accurate than measurement of BP by noninvasive means, especially in the critically ill.

(1) Arterial Pressure Monitoring

- **The purpose of arterial pressure monitoring is to achieve the following points:**

✓ Continuous monitoring of arterial blood pressure,

✓ Tissue perfusion status

✓ Trends in blood pressure

✓ Monitoring of the efficacy of drugs,

✓ Frequent blood samples and arterial blood gas sampling.

(1) Arterial Pressure Monitoring

- The mean arterial pressure (MAP) is the average pressure at which blood moves through the vessels.
- Although true MAP can be calculated only by complex methods (below chart):

Chart 15-1 Formula for Estimating Mean Arterial Pressure (MAP)

$$MAP = \frac{\text{systolic BP} + 2(\text{diastolic BP})}{3}$$

Example: patient's BP = 125/75 mm Hg

$$MAP = \frac{125 + (2 \times 75)}{3}$$

MAP = 92 (rounded to nearest 1/10)

Normal Value of MAP is ranging from (70-100 mmHg)

(1) Arterial Pressure Monitoring

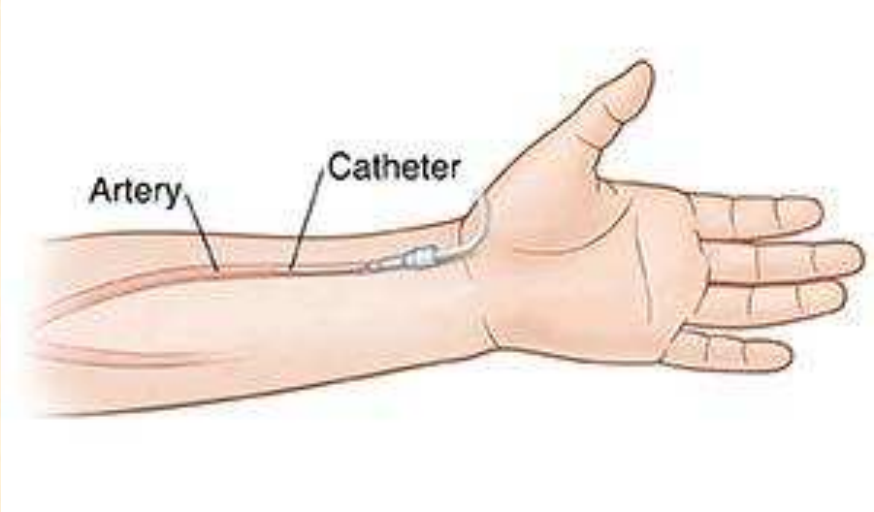
- **Indications for arterial catheterization :**

- Need for continuous blood pressure measurement

- Hemodynamic instability

- Vasopressor requirement

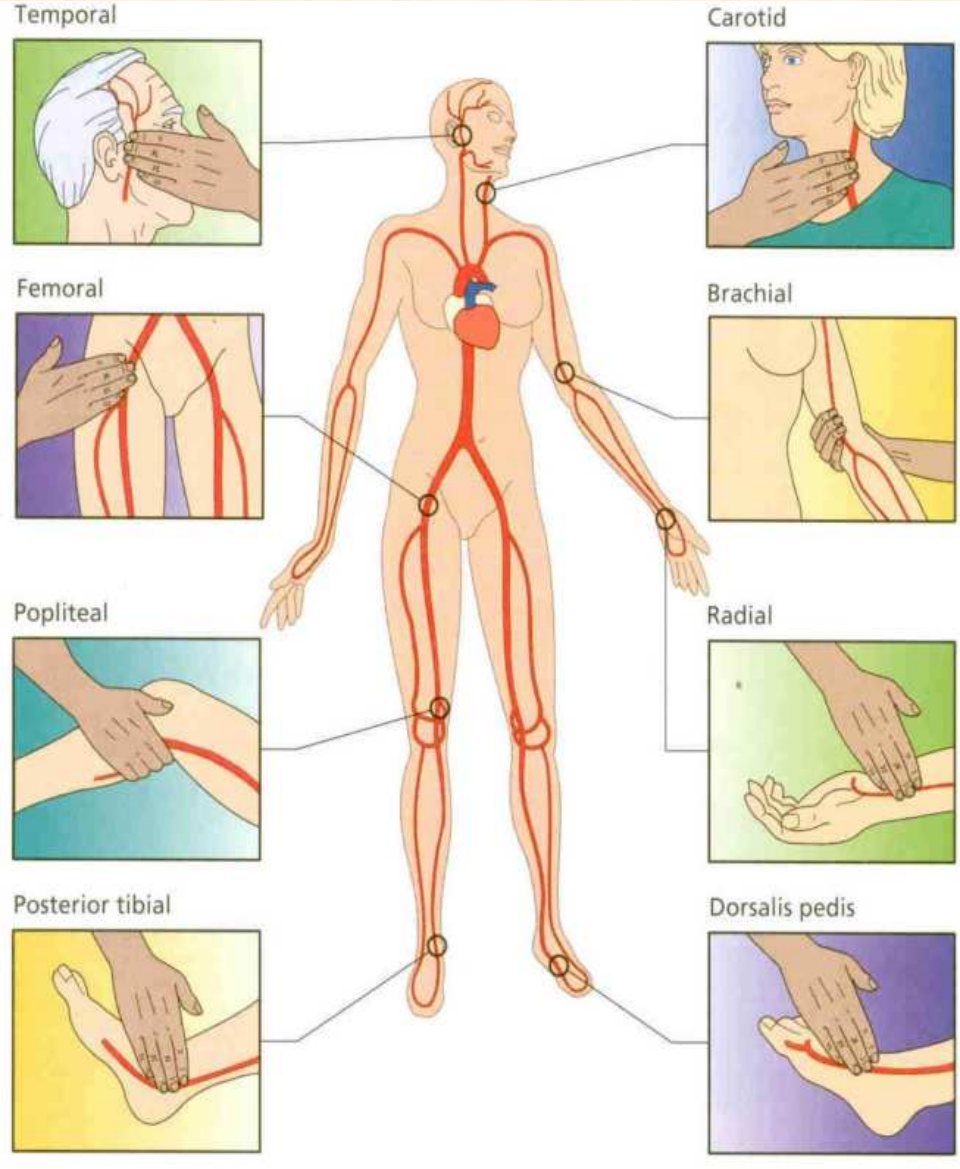
- Respiratory failure



- Frequent Arterial Blood Gas(ABG) Assessments.

(1) Arterial Pressure Monitoring

- **Locations of Arterial Line:**
  - Radial,
  - Ulnar,
  - Brachial,
  - Femoral,
  - Axillary,
  - Posterior tibial, and Dorsalis Pedis.
- After the radial artery, the femoral artery is the second most common site for arterial cannulation.



(1) Arterial Pressure Monitoring

Special Considerations for arterial catheterization:

- **NO medication is Injected in the line.**



- Artery very susceptible to spasm and stop blood flow, so inject only with heparinized saline,
- Never force the line specially if the line appears to be clotted, if the clot forced it may travel on down the limbs and cause blockage of circulation.

(1) Arterial Pressure Monitoring

Special Considerations for arterial catheterization:

- Check all connections frequently: any disconnection may result in rapid loss of a large amount of blood.
- Check the site of insertion include assessing: color, temp., edema, movement, capillary refill, leaking, discomfort and pulse.

(1) Arterial Pressure Monitoring

Complications of arterial catheterization:

- Hemorrhage
- Vascular insufficiency
- Thrombus or emboli
- Infection
- Hematoma
- Ischemia.

(2) Central venous pressure monitoring

(2) Central venous pressure monitoring

- **A Central Venous Catheter (CVC):** is an indwelling intravenous device that is inserted into a vein of the central vasculature (Vessels).
- The CVP, the pressure in the vena cava or right atrium, is used to assess right ventricular function and venous blood return to the right side of the heart.
- The CVP can be continuously measured by connecting either a catheter positioned in the lower third of superior vena cava.

(2) Central venous pressure monitoring

- Because the pressures in the right atrium and right ventricle are equal at the end of diastole, the CVP is also an indirect method of determining right ventricular filling pressure (preload).
- The normal range for CVP is 5-10cm H_2O (2-6mmHg) when taken from the mid-axillary line at the fourth intercostal space. It increases 3 – 5 cm H_2O when patient is being ventilated.

(2) Central venous pressure monitoring

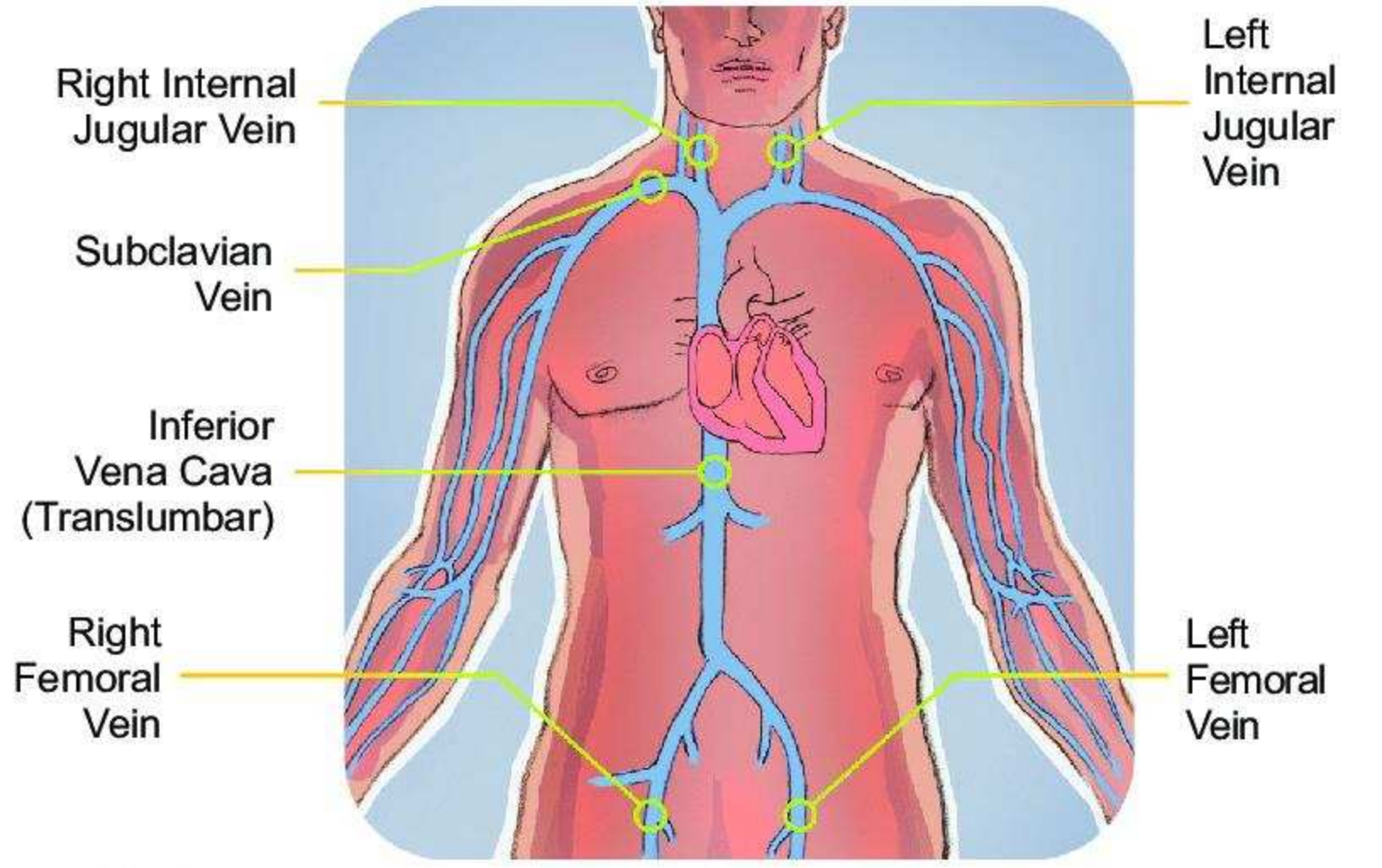
Sites of CVP Line insertion:

There are four common sites:

- Internal jugular,
- Subclavian,
- Femoral, and
- Brachial vein.

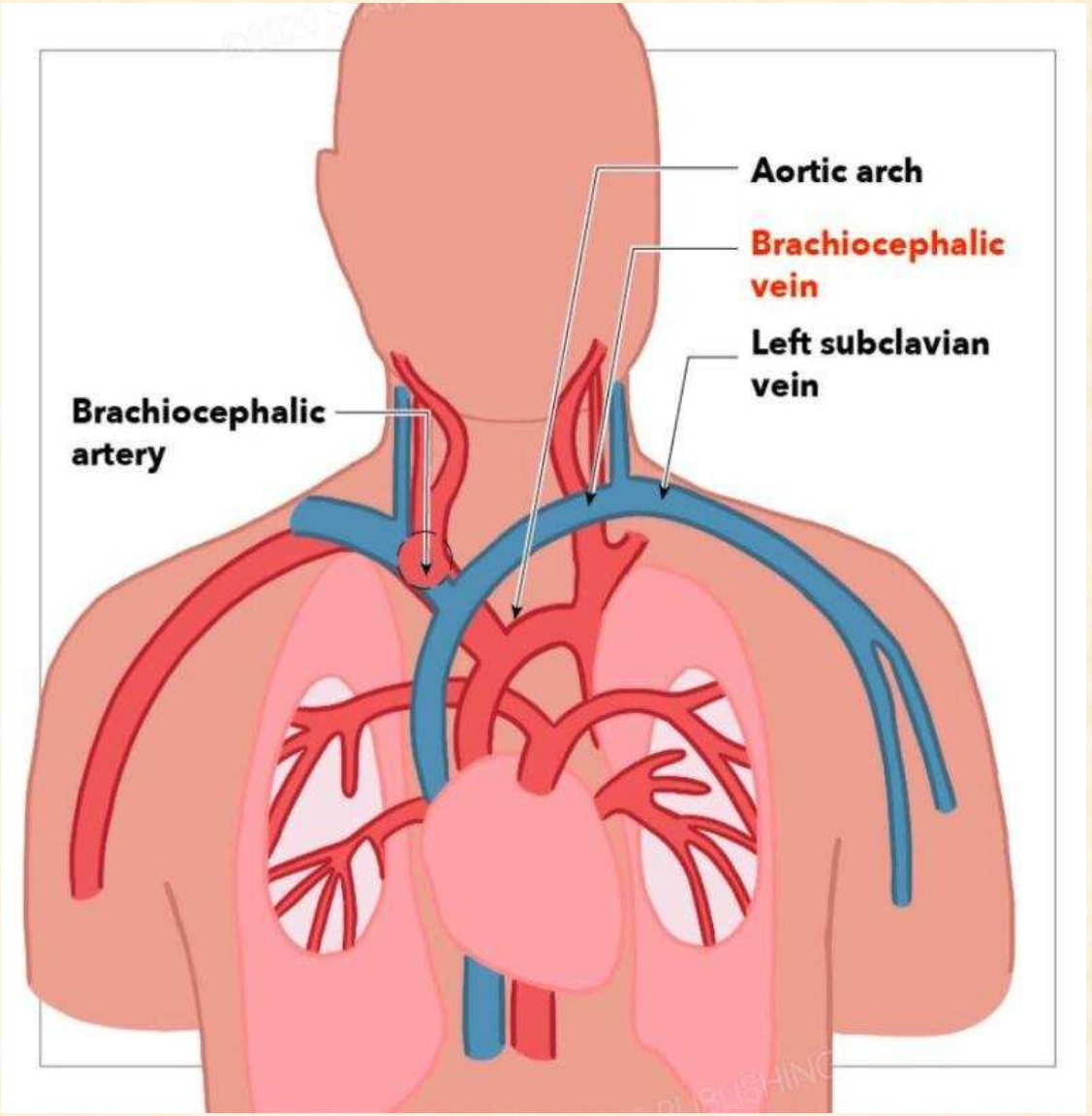
(2) Central venous pressure monitoring

Sites of CVP Line insertion:



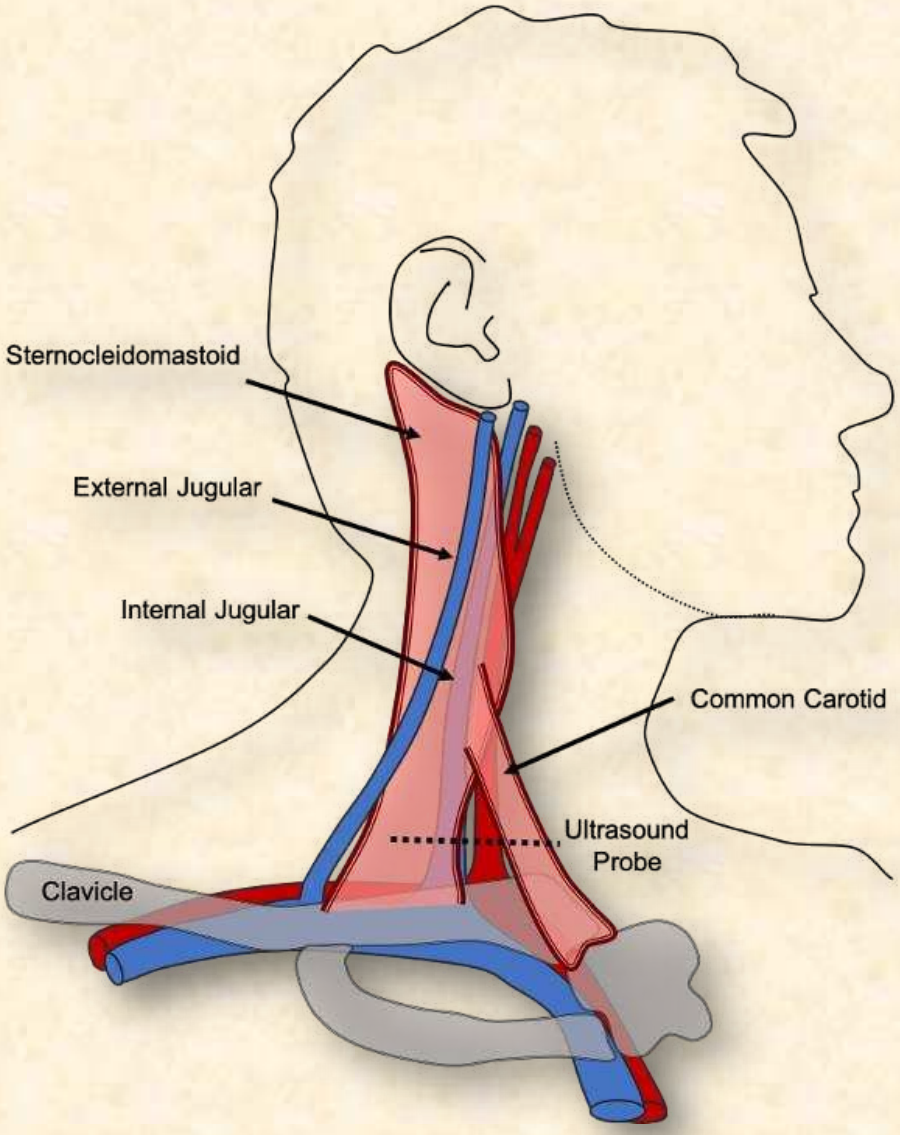
(2) Central venous pressure monitoring

- **Subclavian Vein:**
- **Advantage:**
  - Large vessel
  - Dressing easy
  - Lowest sepsis rate
- **Disadvantage:**
  - Risk of pneumothorax
  - Close to subclavian artery
  - Hard to control bleeding



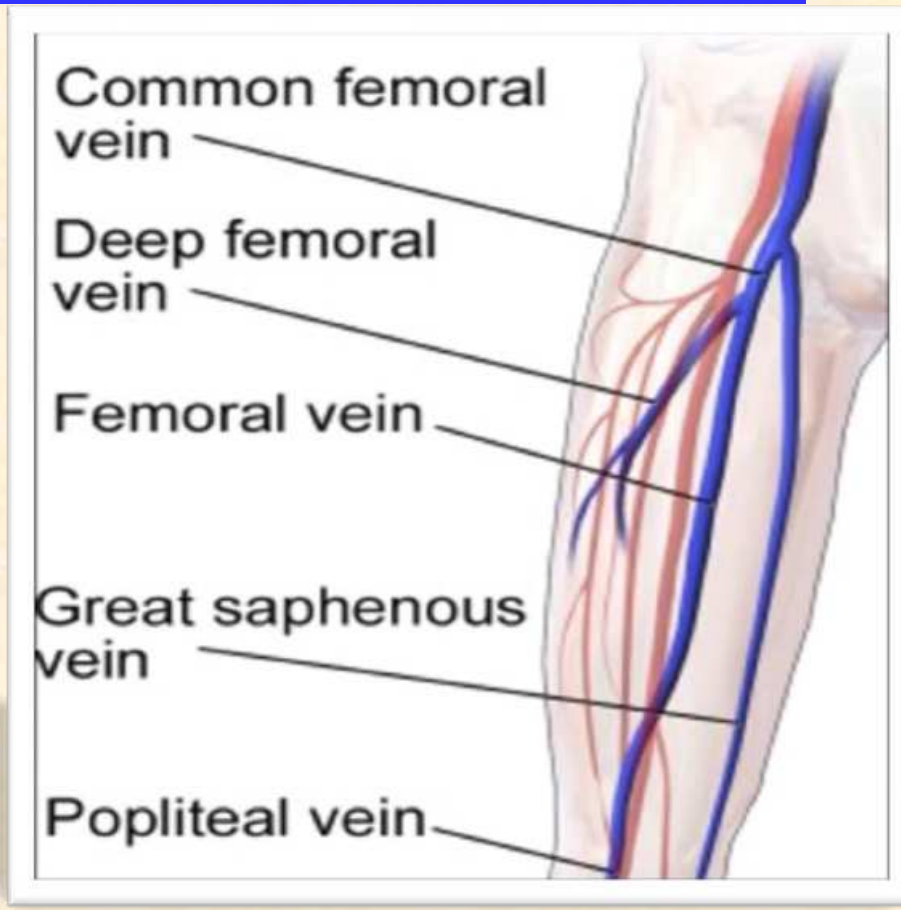
(2) Central venous pressure monitoring

- **Internal Jugular Vein:**
- **Advantage:**
  - Large vessel
  - Easily located and Easy access
  - Decreased risk of pneumothorax
- **Disadvantage:**
  - Uncomfortable to patient
  - Difficult to maintain dressing
  - Difficult maintenance with trachea or neck injury.



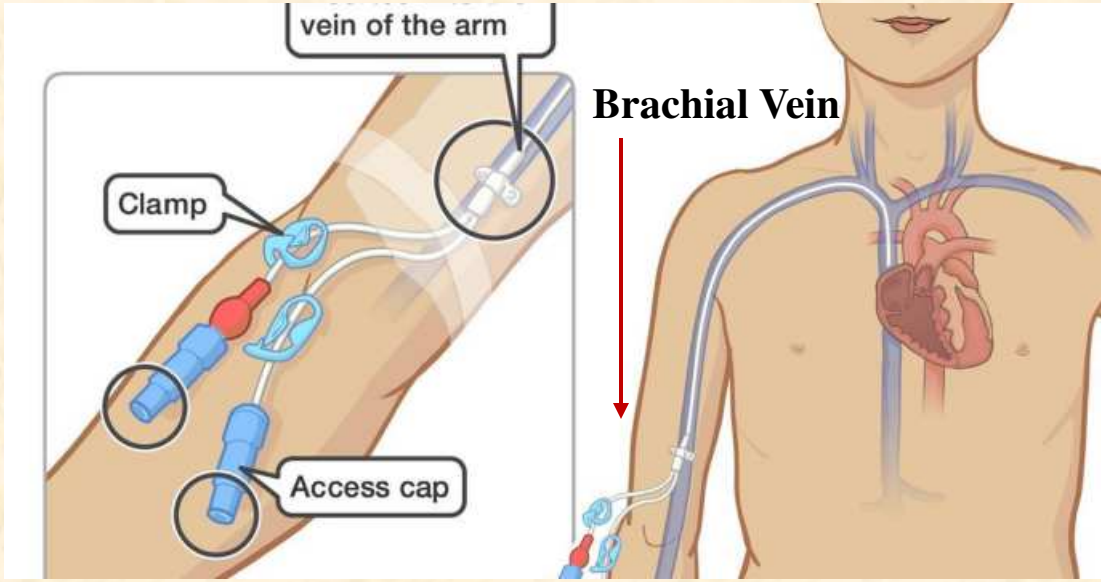
(2) Central venous pressure monitoring

- **Femoral Vein:**
- **Advantage:**
  - Large vessel
  - Easy access
  - Good access during resuscitation.
- **Disadvantage:**
  - Decreased mobility
  - Increased risk of thrombosis, phlebitis & infection & easily contaminated
  - Dressing difficult to maintain.



(2) Central venous pressure monitoring

Brachial Vein:



Advantage:

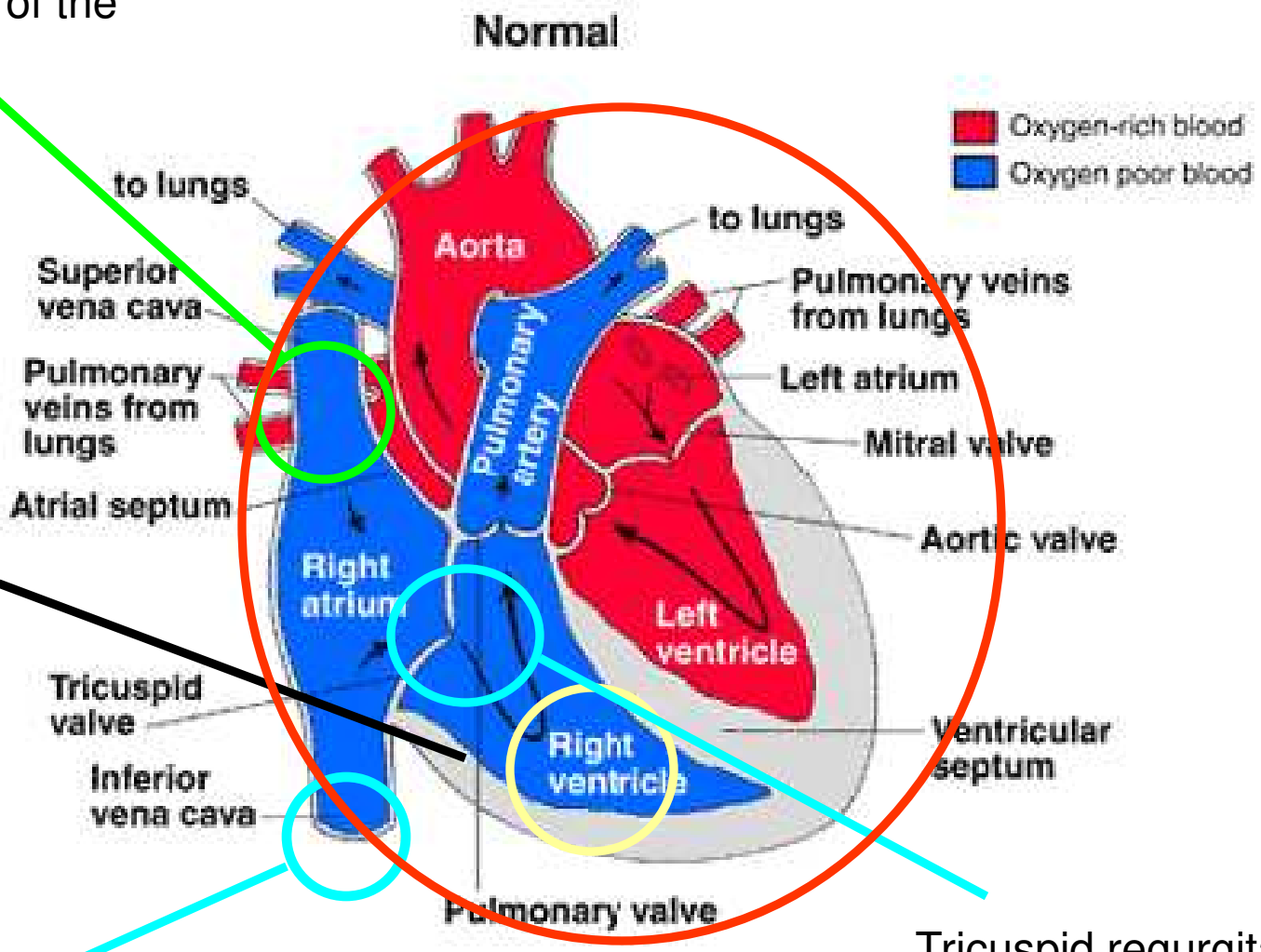
- Large vessel
- Easy access
- Less risk of contamination.

Disadvantage:

- Decreased mobility of the arm
- Increased risk of thrombosis

Limitation of CVP

Obstruction of the great veins



Decrease right ventricular compliance

Tricuspid regurgitation

Systemic venoconstriction

(2) Central venous pressure monitoring

- **Factors affecting Central Venous Pressure (CVP):**
- Systemic vasodilatation and hypovolemia, which leads to reduced venous return in the vena cava.
- Right ventricular failure
- Tricuspid and Pulmonary valve disease
- Pulmonary hypertension

(2) Central venous pressure monitoring

- **Indications for using Central Venous line:**
- Hemodynamic monitoring
- Need for IV access and failure of peripheral access
- Long term IV access anticipated
- Infusion of drugs which may cause peripheral problems e.g. phlebitis

(2) Central venous pressure monitoring

- **Indications for using Central Venous line:**
- Volume resuscitation with large bore central lines
- To obtain frequent blood sample
- To determine how much fluid is needed in 24hrs
- Insertion of a pacing wire.

(2) Central venous pressure monitoring

❑ Central Venous Catheter: Single Lumen vs. Multi-lumen

- ❖ Single, double and triple lumen catheters are available in all catheter types.
- ❖ Each lumen must be treated as a separate catheter.
- ❖ Incompatible medications can be infused simultaneously via separate lumens.
- ❖ When TPN is being infused a lumen MUST be labeled for this use, and nothing else is to be given via that lumen.

Nursing Care for Central venous Catheter Insertion

❑ Pre-Insertion:

- ❖ Prior to insertion ensure the patient/family understands the procedure (Physician responsibility)



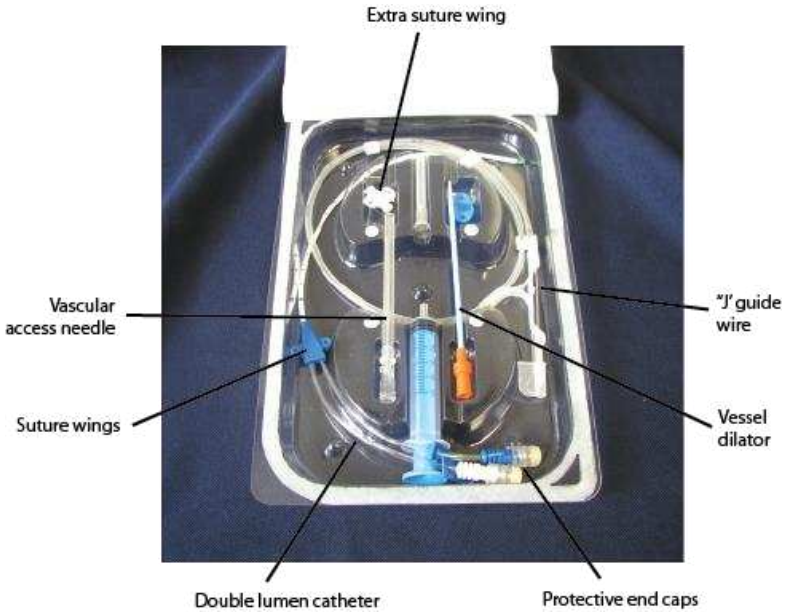
- ❖ Assess patient's vital signs and document.
- ❖ Perform a respiratory assessment including breathing patterns, depth, symmetry, and sounds.
- ❖ The bed is high enough, and Place patient in Trendelenburg position to dilate the veins and reduce the risk of air embolism if tolerated.

Nursing Care for Central venous Catheter Insertion

❑ Pre-Insertion of CVC:

❖ Prepare all equipment needed:

- ✓ Sterile gloves, gown, suture pack
- ✓ Iodine solution
- ✓ 10 ml syringe, 2% lidocaine, 10 ml N.S.
- ✓ Catheter special size
- ✓ H_2O manometer
- ✓ Flush solution with complete CVP line
- ✓ Dressing set.



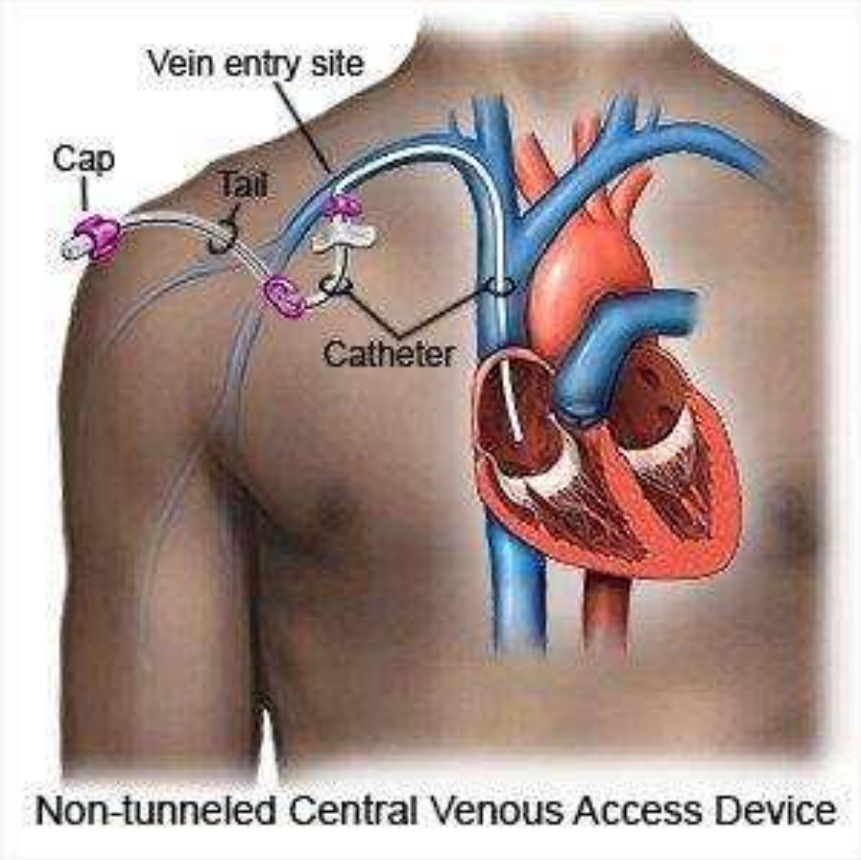


Nursing Care for Central venous Catheter Insertion

❑ The Procedure:

❖ Drape:

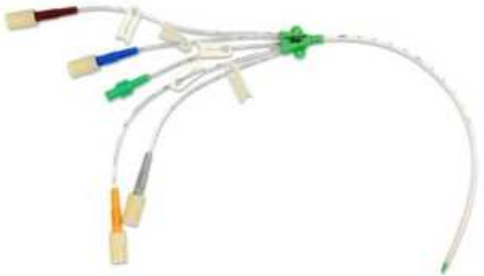
- ○ Large enough
- ○ Handed sterilely by the assistant
- ○ Hole in the area of placement



Non-tunneled Central Venous Access Device

❖ Prepare the tray:

- ○ Handed sterilely by the assistant
- ○ Prepare the equipment before starting



Nursing Care for Central venous Catheter Insertion

❑ Post Insertion of CVC:

- ❖ Order a portable chest x-ray for Physician to confirm correct placement of line.
- ❖ NOTE: Do not use CVC until confirmation of placement.
- ❖ Ensure that all lumens are flushed with 20 mL of NS immediately after insertion.
- ❖ Monitor patient vital signs every 30 min for the first 2 hrs
- ❖ Dispose all sharps
- ❖ Monitor site for bleeding or hematoma.

Nursing Care for Central venous Catheter Insertion

❑ Post Insertion of CVC:

- ❖ Assess breath sounds
- ❖ Cover the line with sterile dressing (prefer transparent dressing).
- ❖ **Documentation to be done on the Central Venous Catheter Insertion including:**

✓ Date & time

✓ Nurse who assisted with the insertion

✓ Physician who inserted the line and their initials

✓ Initial complications

✓ Placement confirmation

(2) Central venous pressure monitoring

❑ Complications associated with Central Venous Catheter:

❖ Air Embolism, Infection, & Occlusion are the three most common complications.

❖ Complications associated with the insertion procedure are: Cardiac dysrhythmias, pneumothorax, hemothorax, bleeding, and hematoma.

❖ **Others complications include**: **Extravasation**, broken catheter tip, arterial puncture and nerve injury.

- **Extravasation** refers to the leakage of injected drugs from blood vessels causing damage to the surrounding tissues.

(2) Central venous pressure monitoring

❑ Central Venous Catheter Displacement:

- If the CVC moves into the chambers of the heart then cardiac arrhythmias may be noted, and should be reported.
- If the CVC is no longer in the correct position, CVP readings and medication administration will be affected.

(3) Pulmonary Artery Catheter

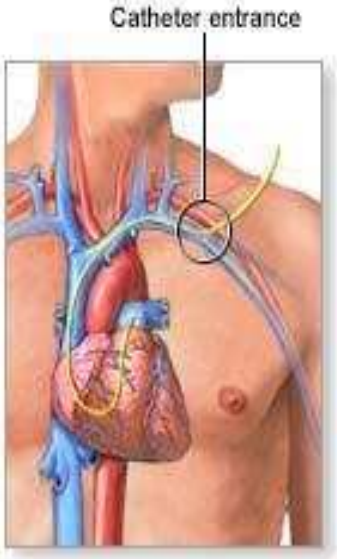
(3) Pulmonary Artery Catheter (Swan-Ganz catheter)

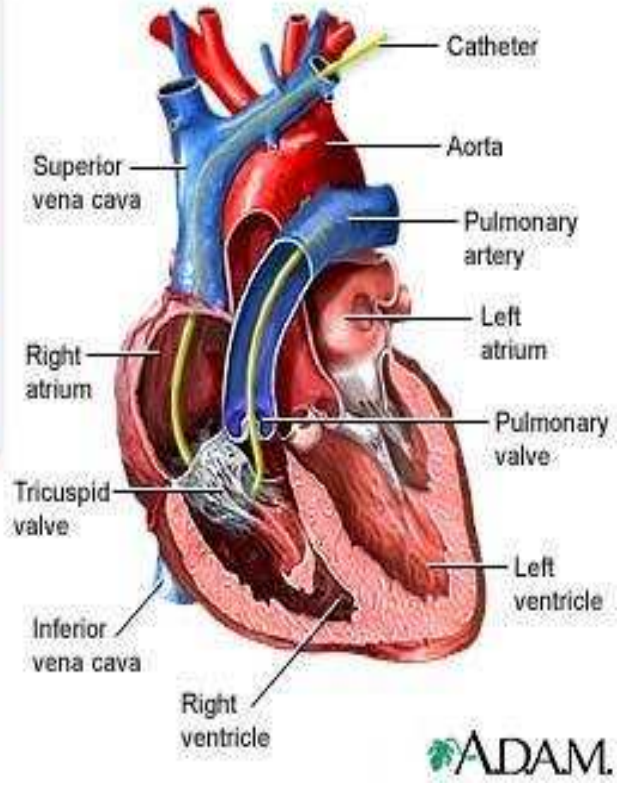
- The Pulmonary Artery catheter (PAC) is an invasive way to obtain hemodynamic measurements by insertion of a catheter into a pulmonary artery.
- It is frequently referred to as a **Swan-Ganz catheter**, in honor of its inventors Jeremy Swan and William Ganz.
- Swan-Ganz catheter for monitoring central venous pressure, pulmonary artery and pulmonary artery wedge pressures (filling pressure of the left atrium).

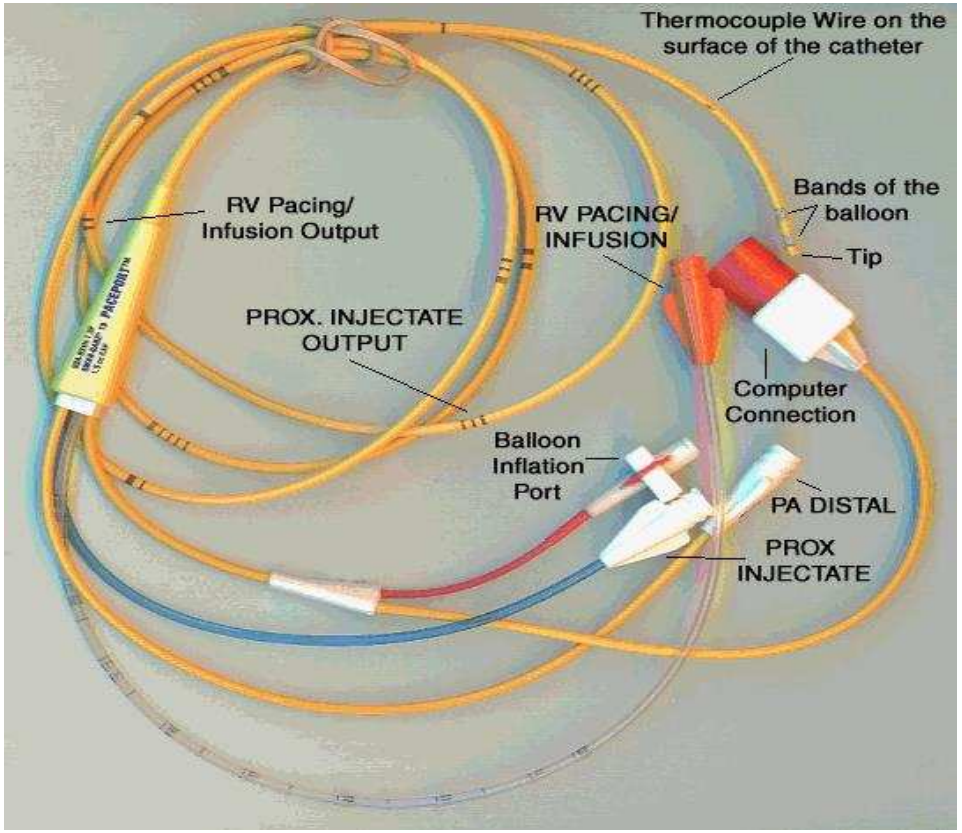
(3) Pulmonary Artery Catheter (Swan-Ganz catheter)

- Can be used for determining cardiac output, for venous and pulmonary artery blood sampling, and for medication administration. Venous lines can be used for fluid administration
- The catheter is introduced through a large vein—often the internal jugular, subclavian, or femoral veins. From this entry site, it is threaded through the right atrium of the heart, the right ventricle, and subsequently into the pulmonary artery. The passage of the catheter may be monitored by dynamic pressure readings from the catheter tip or with the aid of fluoroscopy.

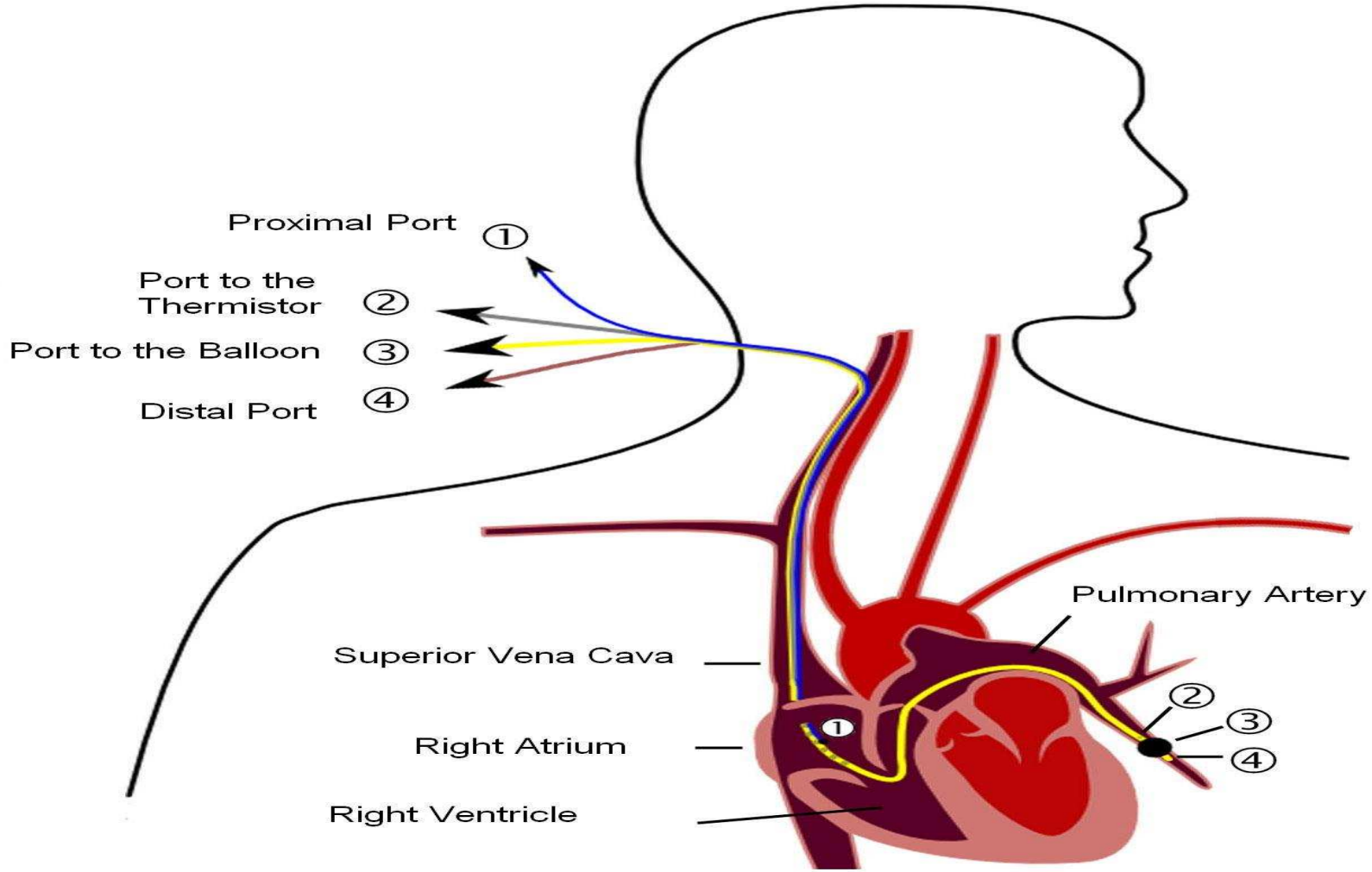
Pulmonary Artery Catheter (Swan-Ganz catheter)







PULMONARY ARTERY Catheter



(3) Pulmonary Artery Catheter (Swan-Ganz catheter)

- Pulmonary capillary wedge pressure (PCWP) measures the pressure generated by the left ventricle, which provides also an indirect estimate of left atrial pressure (LAP).
- Normal pulmonary capillary wedge pressure (PCWP) is 8-12 mmHg at the diastole and 15-25 mmHg at systole, the mean (average of pulmonary artery pressure) is 10-20 mmHg.

(3) Pulmonary Artery Catheter (Swan-Ganz catheter)

❑ Indications for using Pulmonary Artery Catheter:

- ❑ Shock of all types
- ❑ Assessment of cardiovascular function and response to therapy.
- ❑ Assessment of pulmonary status
- ❑ Assessment of fluid requirement
- ❑ Perioperative monitoring.

(3) Pulmonary Artery Catheter (Swan-Ganz catheter)

❑ Clinical Application of Pulmonary Artery Catheter:

❑ PAC can generate large numbers of hemodynamic variables such as:

- Central venous pressure (CVP)
- Pulmonary arterial occlusion pressure (PAOP)
- Cardiac output / cardiac index (CO / CI)
- Stroke volume (SV)
- Right ventricle ejection fraction/ end diastolic volume
- Systemic vascular resistance index (SVRI)
- Pulmonary vascular resistance index (PVRI)
- Oxygen delivery / uptake

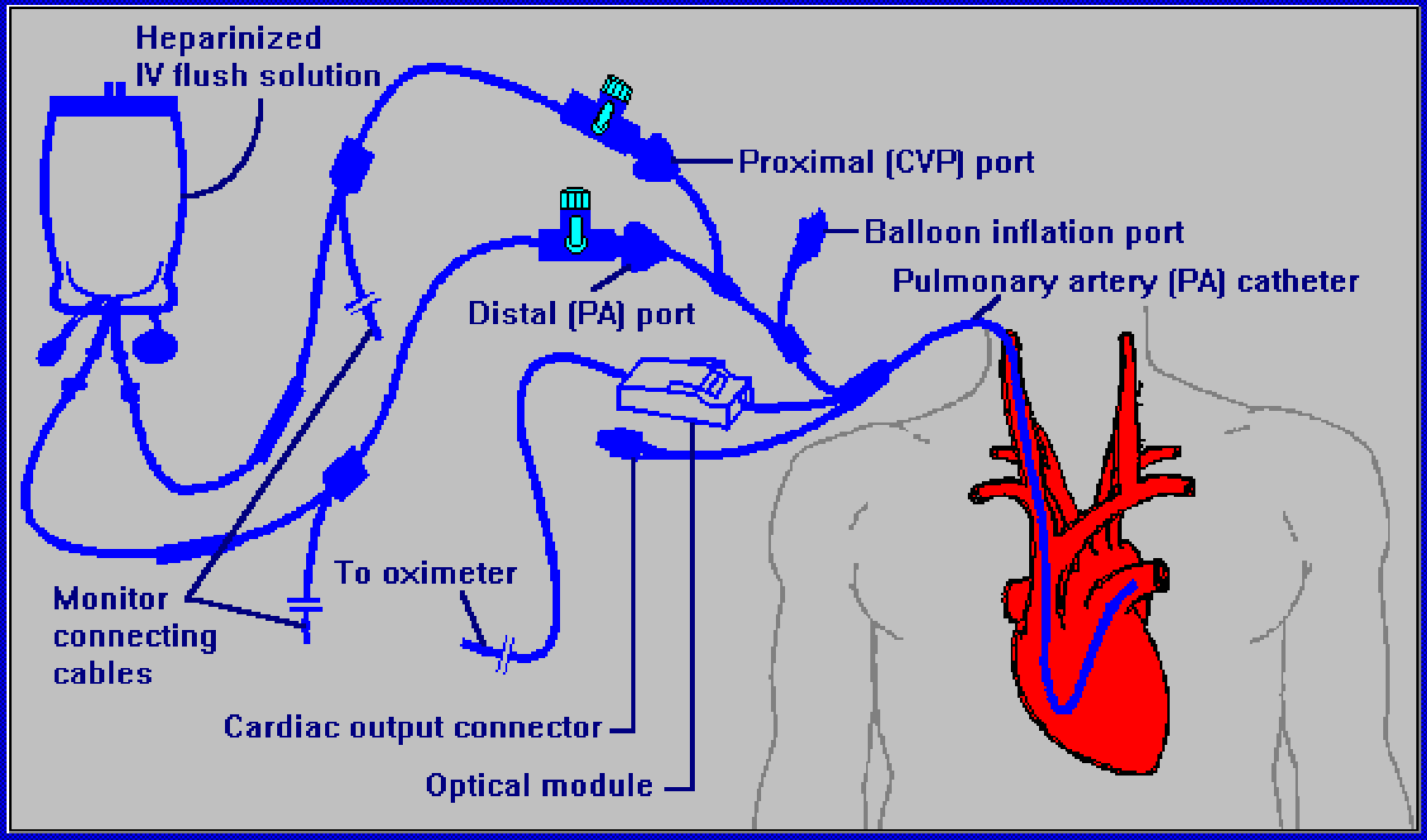
(3) Pulmonary Artery Catheter (Swan-Ganz catheter)

❑ Advantages of Pulmonary Artery Catheter:

- Provide lot of important hemodynamic parameters.
- Sampling site for mixed veinous saturation (SvO_2).

❑ Disadvantages of Pulmonary Artery Catheter:

- Costly
- Invasive
- Multiple complications (eg,, arrhythmia, catheter looping, balloon rupture, PA injury, pulmonary infarction etc..,)
- Mortality is high.



AN EXAMPLE OF A PULMONARY ARTERY PRESSURE MONITORING SYSTEM

Return to Main Menu | Previous | Continue

Measuring hemodynamic parameters

- Before obtaining hemodynamic parameters, the transducer is leveled and zeroed.

1. Explain the procedure to the patient to gain informed consent.

2. If IV fluid is not running, ensure that the CVC is patent by flushing the catheter.

3. Place the patient flat in a supine position if possible.

- Zero reference point is: mid-axillary line at the fourth intercostal space (Phlebostatic axis)



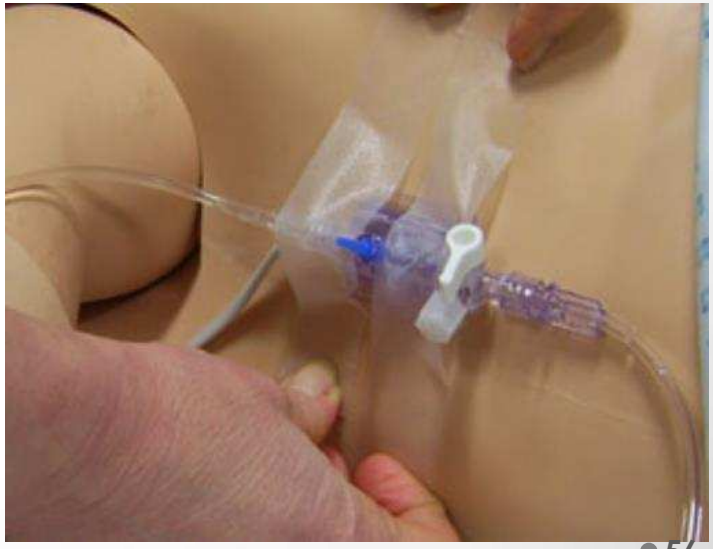
Figure 3: The phlebostatic axis, marked on the patient's chest, is the precise anatomical point of origin of the hemodynamic pressures being measured.

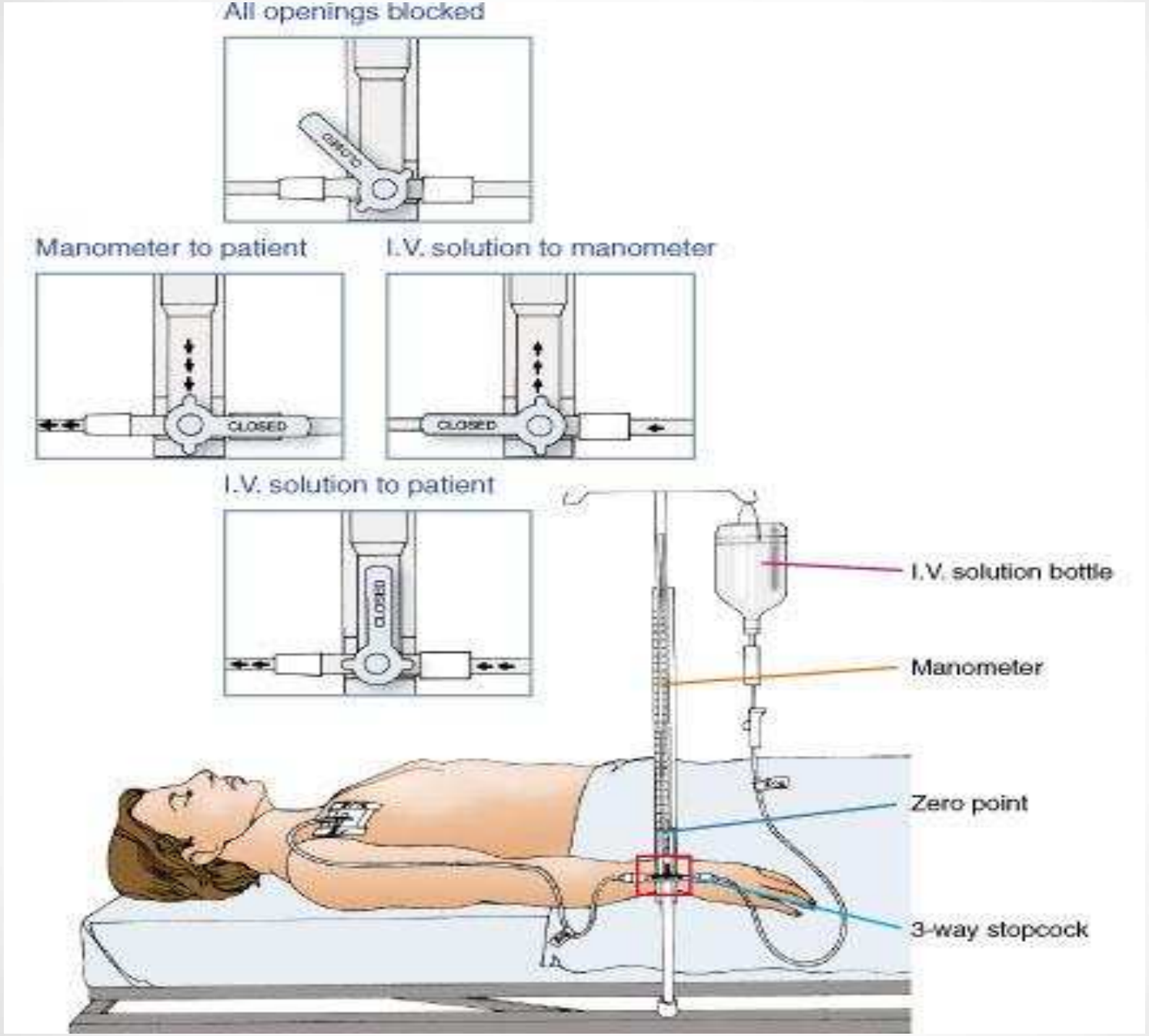
Methods

1. **Using a manometer**



2. **Using a transducer**



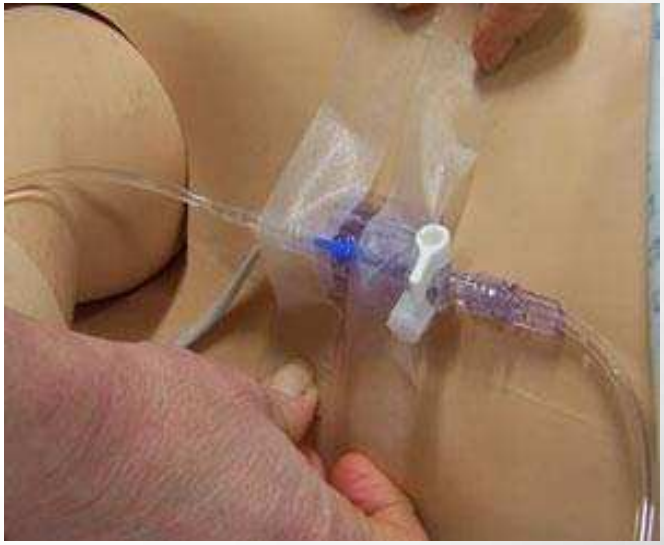

All openings blocked
Manometer to patient
I.V. solution to manometer
CLOSED
CLOSED
I.V. solution to patient
CLOSED
I.V. solution bottle
Manometer
Zero point
3-way stopcock

Using a transducer

1. white or proximal lumen is suitable for measuring CVP.



2. Tape the transducer to the phlebostatic axis or as near to the right atrium as possible

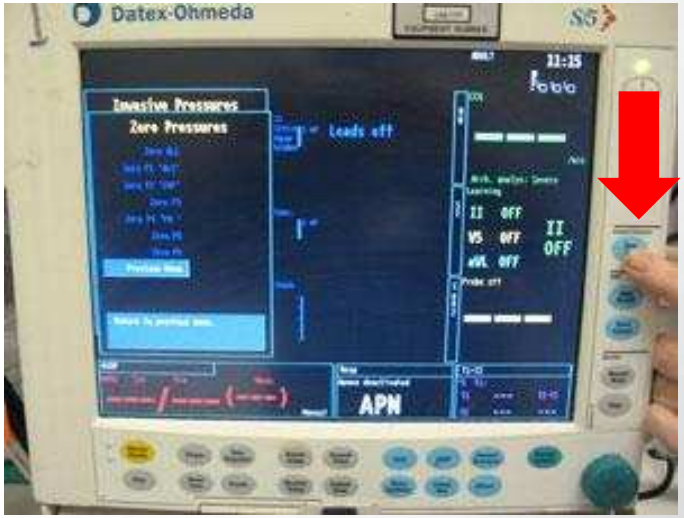


Continue

3. Turn the tap off to the patient and open to the air by removing the cap from the three-way port opening the system to the atmosphere.



4. Press the zero button on the monitor and wait while calibration occurs.



Continue

5. When 'zeroed' is displayed on the monitor, replace the cap on the three-way tap and turn the tap on to the patient.



6. Observe the CVP trace on the monitor. The waveform undulates as the right atrium contracts and relaxes, emptying and filling with blood. (light blue in this image)



Normal value

Parameter	Normal value
Mean Arterial Pressure (MAP)	70 -90 mm Hg
Cardiac Index (CI)	2.8-4.2 L/min/m2
Cardiac Output (CO)	4-8 L/min
Central Venous Pressure (CVP)	2-6 mmHg
Atrial Pressure (RA))	2-8mm H2O
Pulmonary Artery Pressure (PA)	Systolic 20-30 mmHg (PAS) Diastolic 8-12 mmHg (PAD) Mean 15-25 mmHg
Pulmonary Capillary Wedge Pressure (PWCP)	8-12 mmHg

Nursing Interventions

- Once catheter is inserted, it is secured and a dry, sterile dressing is applied.
- Catheter placement is confirmed by a chest x-ray, and the site is inspected daily for signs of infection.
- The dressing and pressure monitoring system or water manometer are changed according to hospital policy.
- In general, the dressing is to be kept dry and air occlusive.
- Dressing changes are performed with the use of sterile technique.

Nursing Interventions

- CVP catheters can be used for infusing intravenous fluids, administering intravenous medications, and drawing blood specimens in addition to monitoring pressure.
- The transducer must be positioned at the phlebo-static axis to ensure accurate readings.
- The nurse who obtains the wedge reading ensures that the catheter has returned to its normal position in the pulmonary artery by evaluating the pulmonary artery pressure waveform.
- Critically ill patients usually require higher left ventricular filling pressures to optimize cardiac output.

Nursing Responsibilities

Site Care & Catheter Safety:

- A sterile dressing is placed over the insertion site and the catheter is taped in place.
- The insertion site should be assessed for infection and the dressing changed every 72 hours and PRN.
- The placement of the catheter, stated in centimeters, should be documented and assessed every shift.

Nursing Responsibilities

Site Care & Catheter Safety:

- The integrity of the sterile sleeve must be maintained so the catheter can be pulled back without contamination.
- The catheter tubing should be labeled and all the connections secure.
- The balloon should always be deflated and the syringe closed and locked unless you are taking a hemodynamic measurement.

Nursing Responsibilities

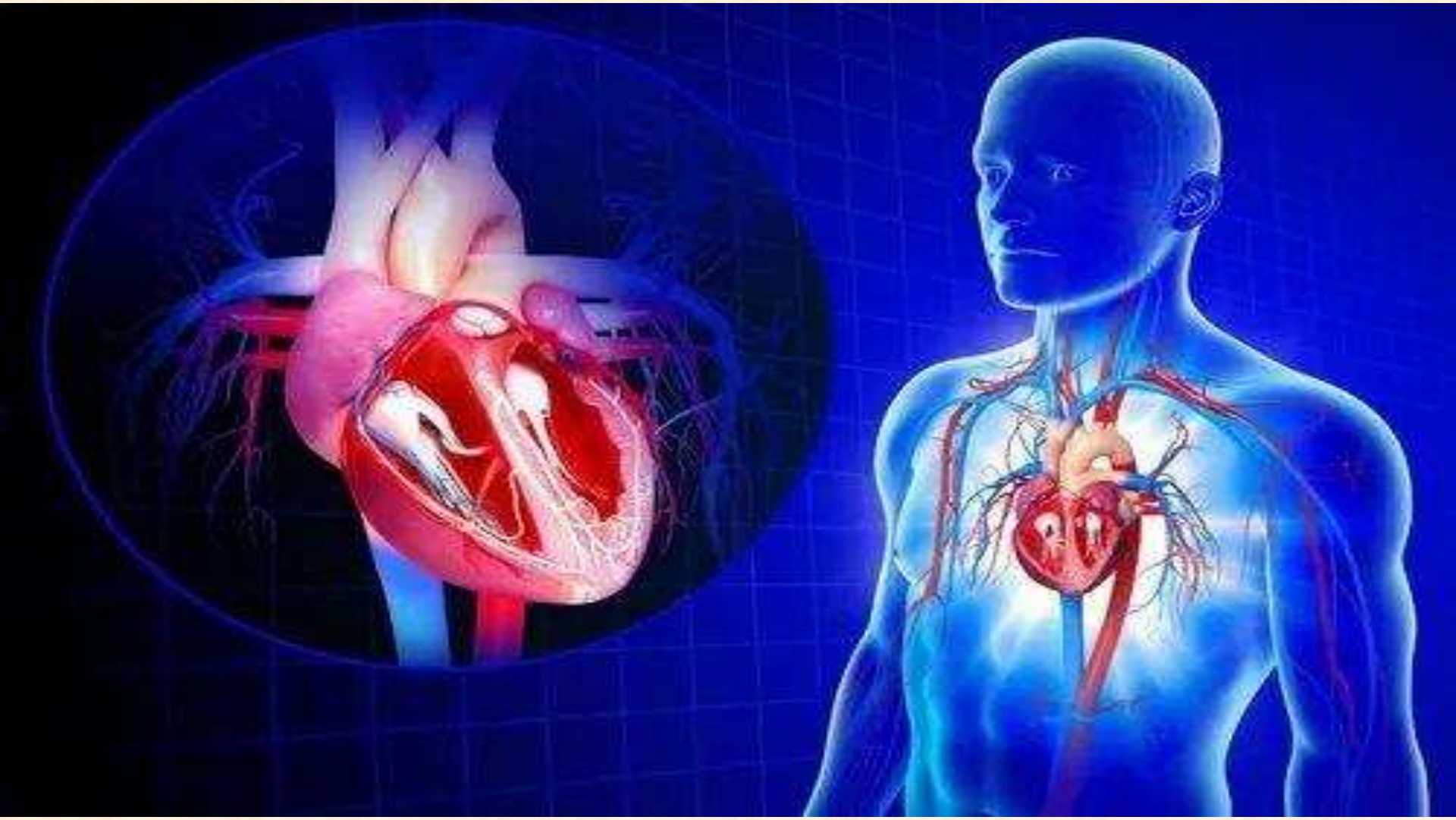
❑ Patient Activity & Positioning:

- Many physicians allow stable patients who have PA catheters, such as post CABG patients, to get out of bed and sit. The nurse must position the patient in a manner that avoids dislodging the catheter.
- Proper positioning during hemodynamic readings will ensure accuracy.



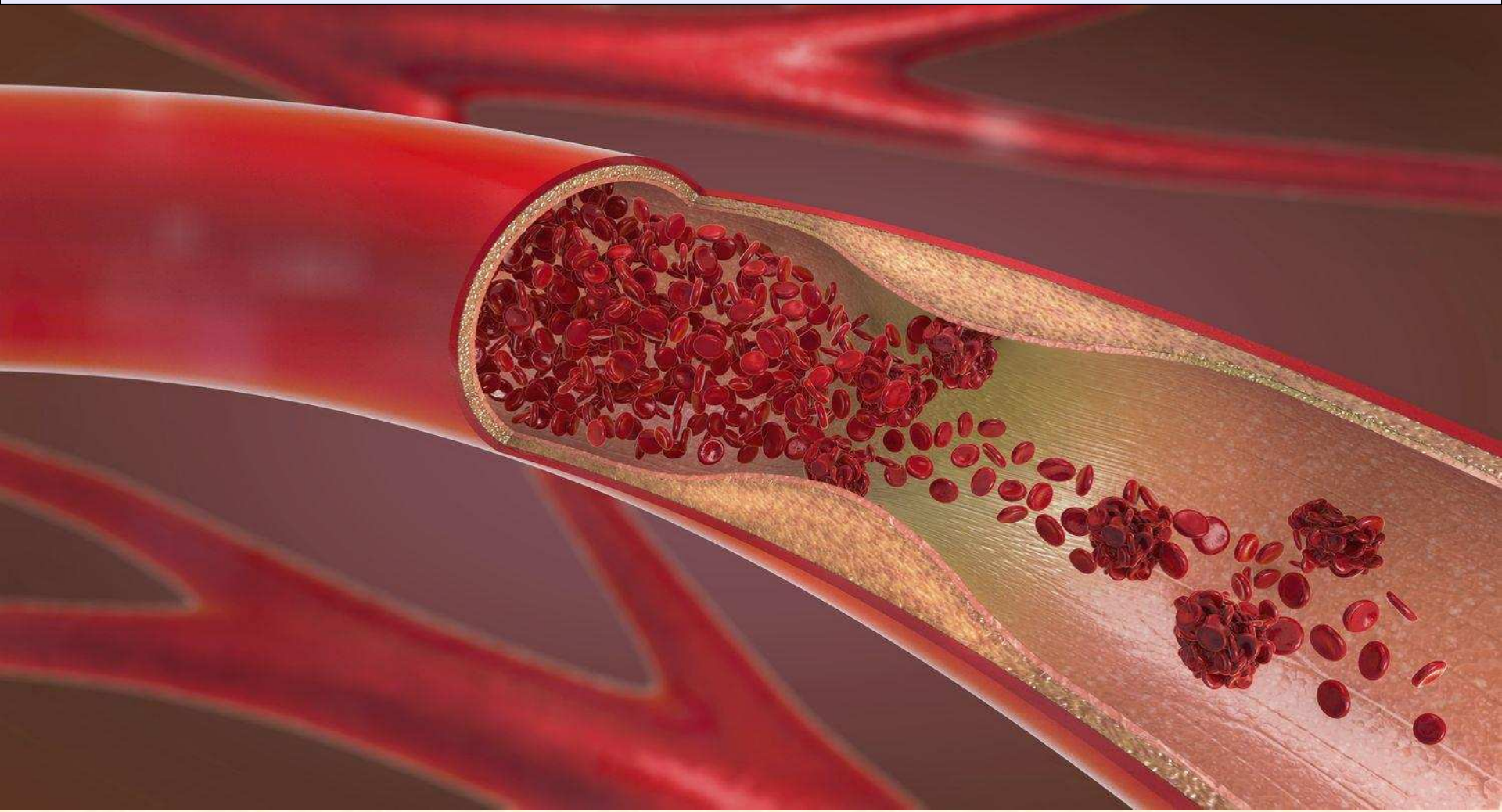
QUESTIONS?

Vascular Disorders and Management



BY: Dr. Wael S. Mekky

Vascular Disorders (Lecture 9)



BY: Dr. Wael S. Mekky

Arteriosclerosis and Atherosclerosis

Arteriosclerosis and Atherosclerosis

- Arteriosclerosis and Atherosclerosis are sometimes used to mean the same thing, but there's a difference between the two terms.

- Arteriosclerosis occurs when the blood vessels that carry oxygen and nutrients from the heart to the (arteries) become thick and stiff — sometimes restricting blood flow to the organs and tissues. Healthy arteries are flexible and elastic. But over time, the walls in the arteries can harden, a condition commonly called hardening of the arteries.

Arteriosclerosis and Atherosclerosis

- **Atherosclerosis** is a specific type of **Arteriosclerosis.**
- The most common peripheral vascular disease is atherosclerosis, which characterized by an abnormal accumulation of lipid, or fatty substances and fibrous tissue in the vessel wall. These accumulations are referred to as atheroma or plaques ترسبات.
- In vascular atherosclerosis, blockages and narrowing of the vessels and thus reduce blood flow to vital organs and elevate arterial blood pressure.

Atherosclerosis

- Atherosclerosis is the buildup of fats, cholesterol and other substances in and on the artery walls. This buildup is called plaque. The plaque can cause arteries to narrow, blocking blood flow. The plaque can also burst, leading to a blood clot.
- Although atherosclerosis is often considered a heart problem, it can affect arteries anywhere in the body.
- Atherosclerosis can be treated.
- Healthy lifestyle habits can help prevent atherosclerosis.

Atherosclerosis

Symptoms of Atherosclerosis:

- Mild Atherosclerosis usually doesn't have any symptoms.
- Atherosclerosis symptoms usually don't happen until an artery is so narrowed or clogged انسداد that it can't supply enough blood to organs and tissues. Sometimes a blood clot completely blocks blood flow. The clot may break apart and can trigger a heart attack or stroke.

Atherosclerosis

Symptoms of Atherosclerosis:

- Symptoms of moderate to severe atherosclerosis depend on which arteries are affected. For example:
- If the atherosclerosis in heart arteries: chest pain or pressure (angina).
- If the atherosclerosis in the arteries leading to brain: sudden numbness or weakness in the arms or legs, difficulty speaking or slurred speech, temporary loss of vision in one eye, or drooping muscles in face (Signs of Transient Ischemic Attack "TIA"). If not treated leads to Stroke.

Atherosclerosis

Symptoms of Atherosclerosis:

- If the atherosclerosis in the arteries in the arms and legs: symptoms of peripheral artery disease, such as leg pain when walking or decreased blood pressure in an affected limb.
- If the atherosclerosis in the arteries leading kidneys: develop of high blood pressure or kidney failure.
- NOTE: Early diagnosis and treatment can stop atherosclerosis from worsening and prevent a heart attack, stroke or another medical emergency.

Atherosclerosis

❑ Causes of Atherosclerosis:

❖ Atherosclerosis is a slowly worsening disease that may begin as early as childhood. The exact cause is unknown. It may start with damage or injury to the inner layer of an artery. The damage may be caused by:

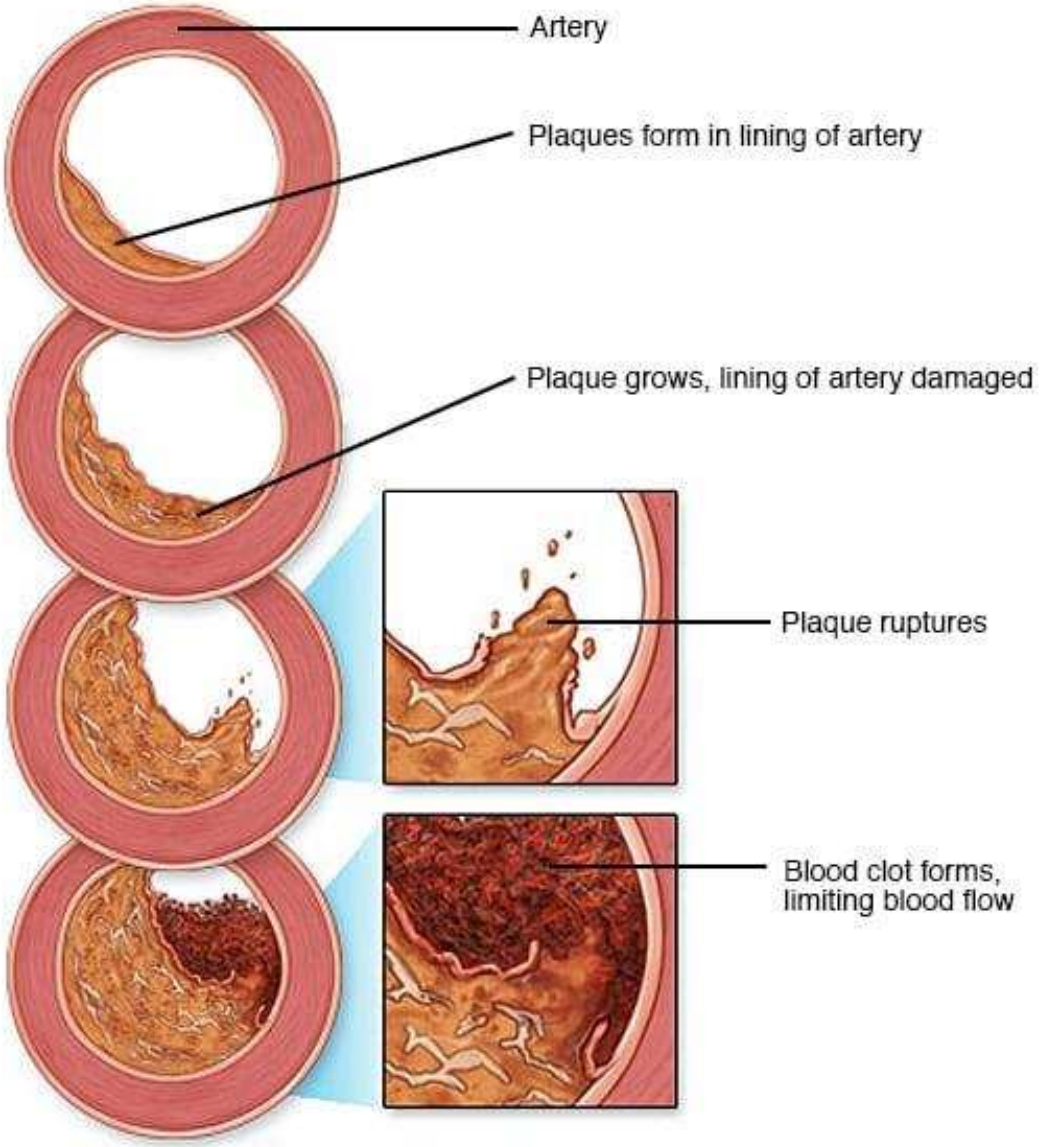
- High blood pressure and Smoking
- High cholesterol, and High triglycerides
- Diabetes and Insulin resistance
- Obesity
- Inflammation from an unknown cause or from diseases such as arthritis, inflammatory bowel disease.

Atherosclerosis

Causes of Atherosclerosis:

- Once the inner wall of an artery is damaged, blood cells and other substances may gather at the injury site and build up in the inner lining of the artery.
- Over time, fats, cholesterols and other substances also collect on the inner walls of the heart arteries. This buildup is called plaque.
- Plaque can cause the arteries to narrow, blocking blood flow. The plaque can also burst, leading to a blood clot.

Development of Atherosclerosis



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Atherosclerosis

Risk Factors of Atherosclerosis:

- Hardening of the arteries occurs over time.
- Aging is a risk factor for Atherosclerosis.
- Other things that may increase the risk of Atherosclerosis include:

- A family history of early heart disease
- An unhealthy diet
- Diabetes

Atherosclerosis

❑ Risk Factors of Atherosclerosis:

- High blood pressure
- High cholesterol
- High levels of C-reactive protein (CRP), a marker of inflammation
- Lack of exercise
- Obesity
- Sleep apnea
- Smoking and other tobacco use

Atherosclerosis

Complications of Atherosclerosis:

- The complications of Atherosclerosis depend on which arteries are narrowed or blocked. For example:
- Coronary Artery Disease: which can cause chest pain (angina), a heart attack or heart failure.
- Carotid Artery Disease: This can cause a transient ischemic attack (TIA) or stroke.
- Peripheral artery disease: a lack of blood flow to the arms or legs can cause tissue death (gangrene).

Atherosclerosis

Complications of Atherosclerosis:

- **Aneurysms:** a serious complication that can occur anywhere in the body. Most people with aneurysms have no symptoms. Pain in the area of an aneurysm may occur and is a medical emergency. If an aneurysm occurs, it can cause life-threatening bleeding inside the body.

- **Chronic Kidney Disease:** Atherosclerosis can cause the arteries leading to the kidneys to narrow. Narrowing of these arteries prevents enough oxygen-rich blood from reaching the kidneys.

Atherosclerosis

❑ Prevention of Atherosclerosis:

❖ The healthy lifestyle changes are recommended to treat atherosclerosis also help prevent it. These lifestyle changes can help keep the arteries healthy:

- Quitting smoking
- Eating healthy foods
- Exercising regularly
- Maintaining a healthy weight
- Checking and maintaining a healthy blood pressure
- Checking and maintaining healthy cholesterol and blood sugar levels.

Atherosclerosis

- Medical Management:

- The traditional medical management of atherosclerosis involves:

- Modification of risk factors.

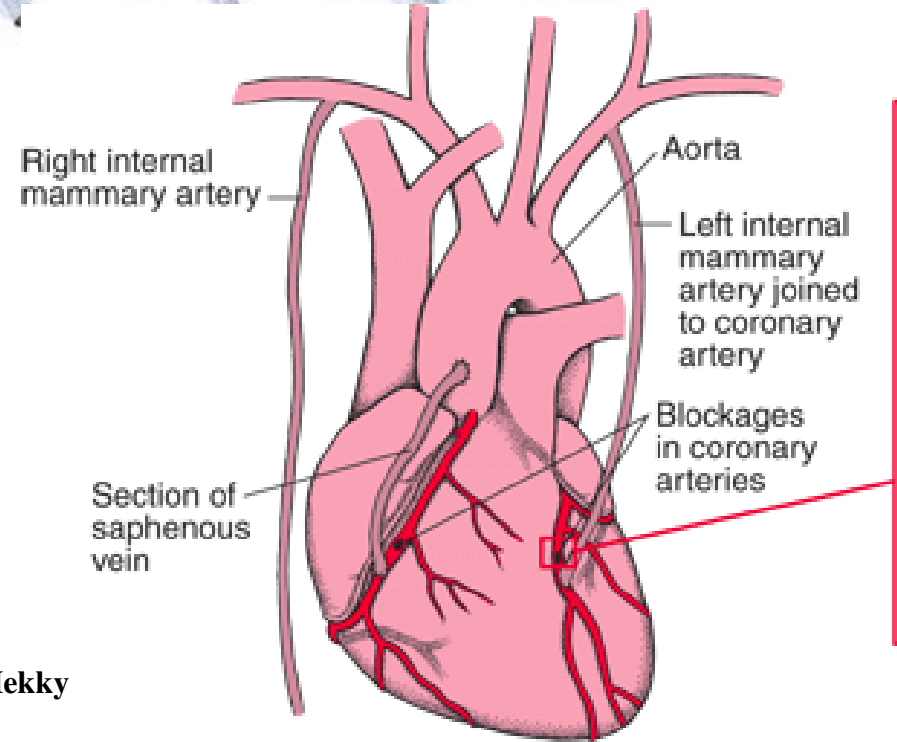
- Exercise to improve circulation and increase circulatory functioning capacity.

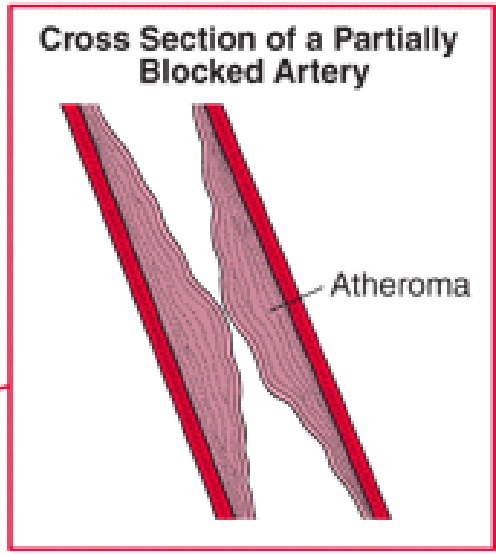
- Medications (Reduce blood lipid levels) and use of antiplatelet agents.

- **Surgical graft procedures.**

Invasive treatment: angioplasty, stenting and bypass grafting

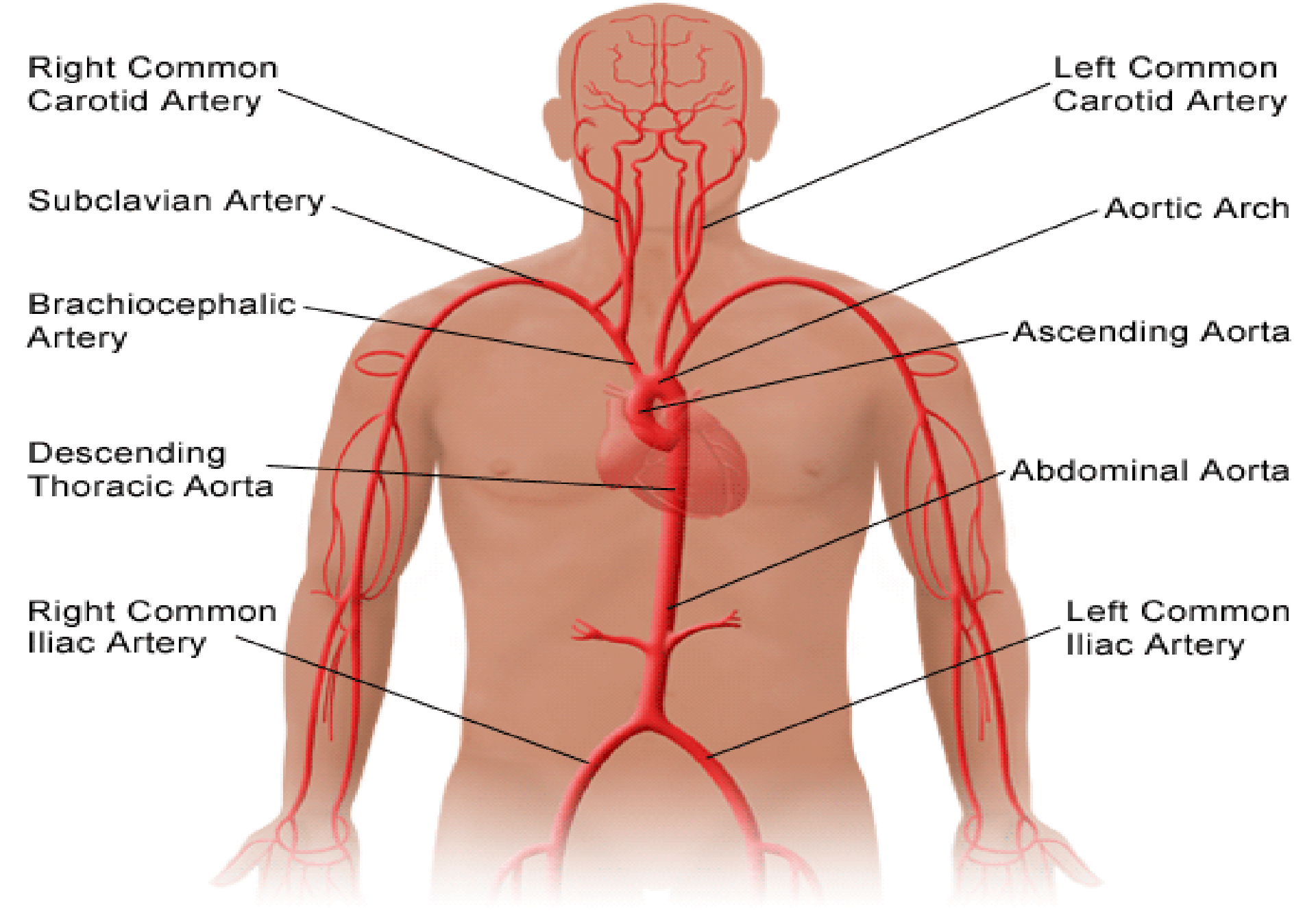






Aortic Aneurysms

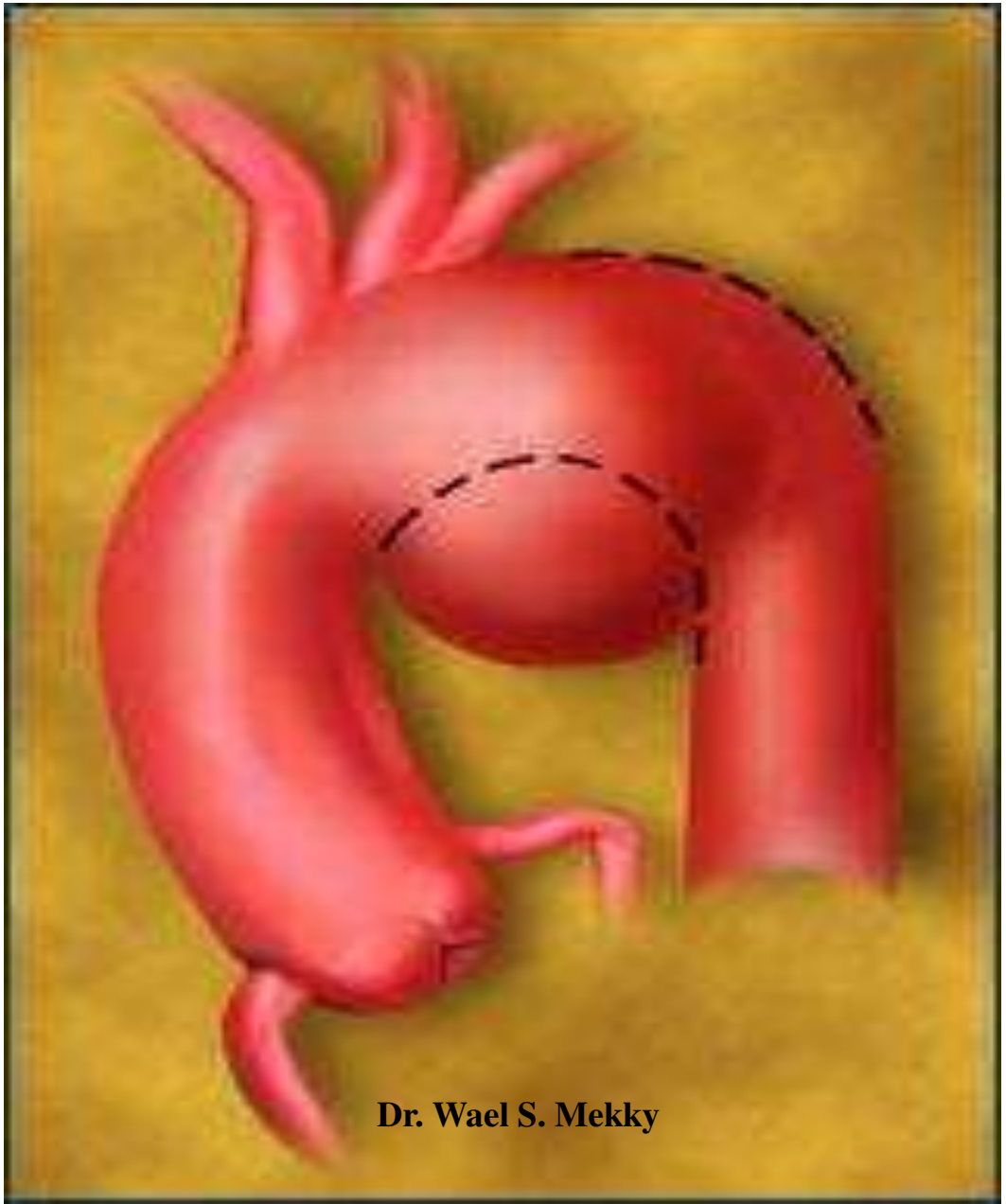
Anatomy of the Aorta



Aortic Aneurysms

- An aneurysm is defined as a segmental, full-thickness dilation of a blood vessel 50 percent greater than its normal diameter.
- Aortic aneurysm is a localized sac or dilation formed at a weak point in the wall of the aorta; that the normal diameter of aorta varies with age, gender, and body habitus.
- It is cause weakness in the wall of the aorta and increases the risk of aortic rupture.
- When rupture occurs, massive internal bleeding results and, unless treated immediately, shock and death can occur.

Aortic Aneurysms



Aortic Aneurysms

Causes of Aortic aneurysm:

- Most aneurysms are arteriosclerotic in origin
- Syphilis
- Infection or Inflammatory diseases
- Trauma
- Hypertension
- Smoking
- Aortic dissection

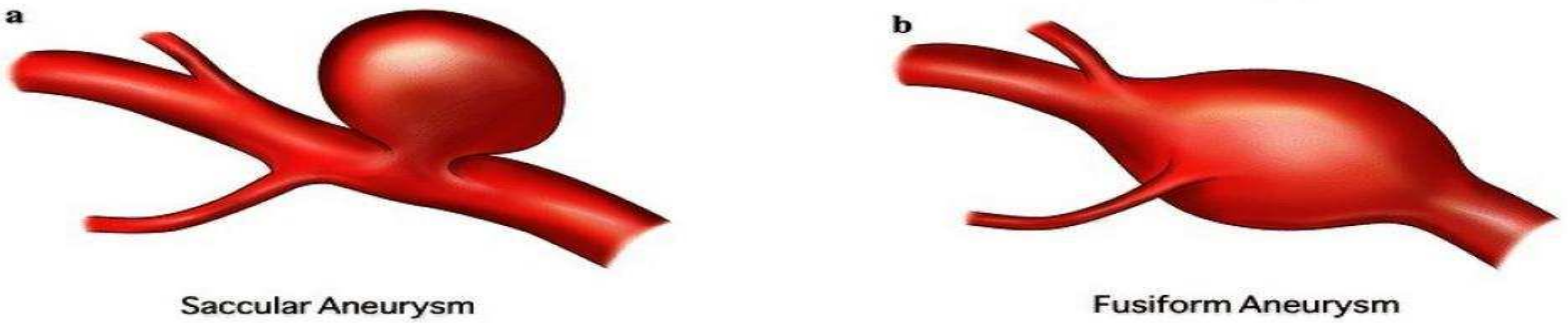
Aortic Aneurysms

- Classifications of Aortic Aneurysms:

- Aortic aneurysms may be classified into several groups according to their morphology, type or location.

1. Classifications of Aortic Aneurysms according to their shape or morphology:

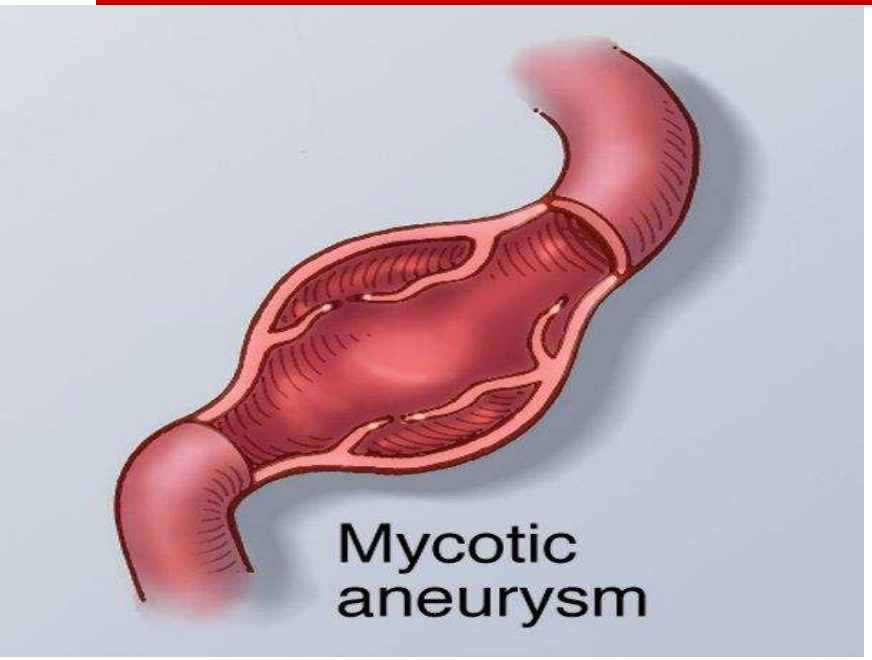
Aortic Aneurysms

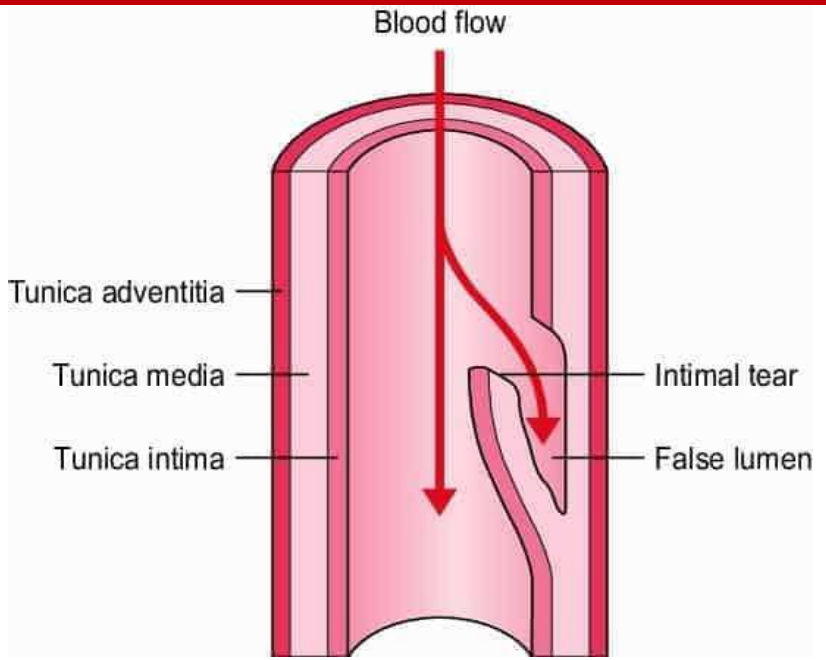


a) Saccular Aneurysm: localized balloon- shaped outpouching projects from one side of the artery.

b) Fusiform Aneurysm: dilation of the entire circumference of the artery.

Aortic Aneurysms





c) Mycotic aneurysms: very small aneurysms due to localized infection (inflammatory digestion of the vessel wall).

d) Dissecting aneurysm: this usually is a hematoma that splits the layers of the arterial wall (between inner and middle layers) due to arteriosclerosis.

Aortic Aneurysms

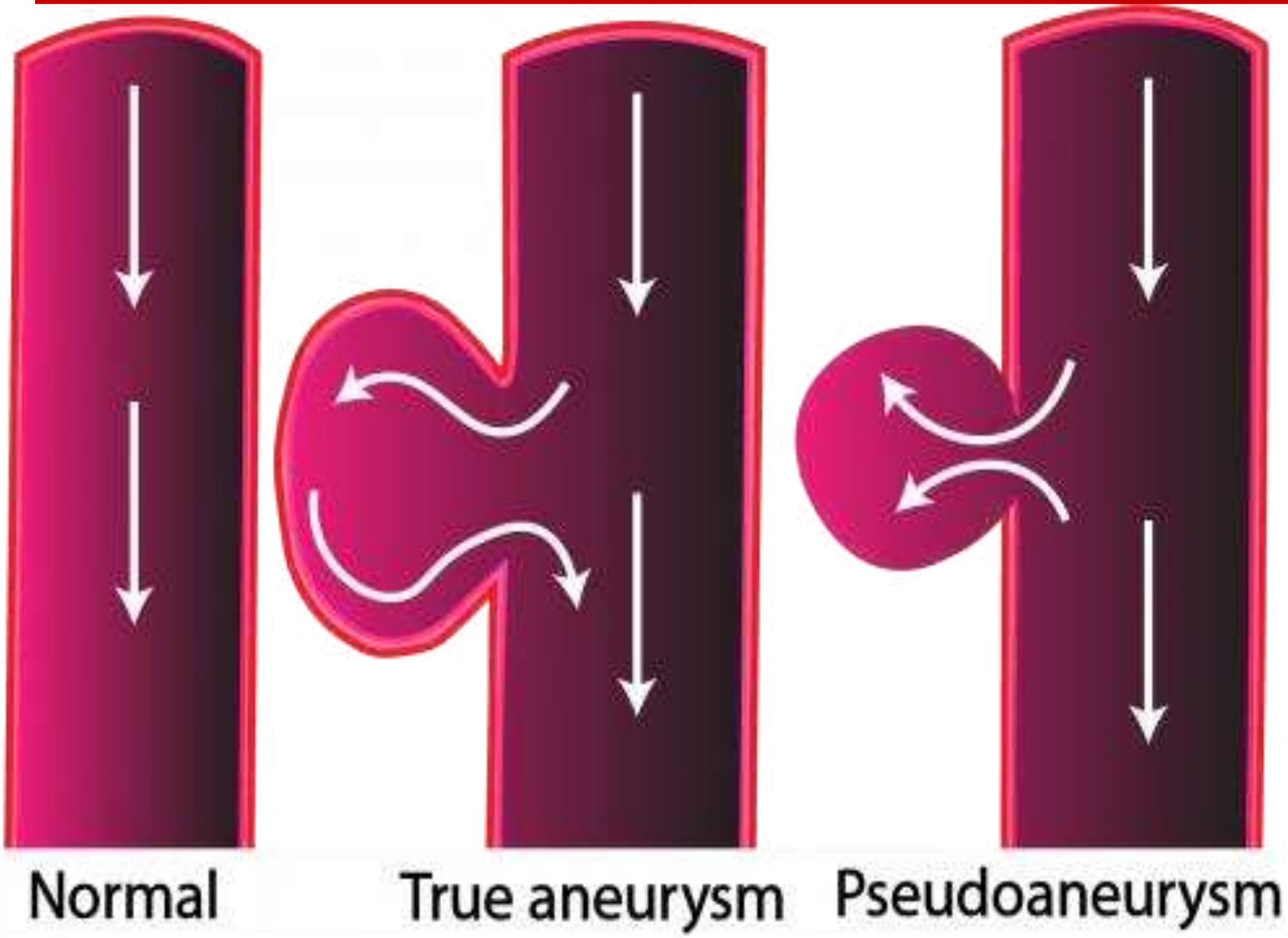
2. Classifications of Aortic Aneurysms according to their type:

I. True Aneurysm: involve the entire vessel wall.

II. False Aneurysm: is formed when blood leaks outside of the artery but is contained by the surrounding tissues.

- A pseudoaneurysm, or false aneurysm, is an enlargement of only the outer layer of the blood vessel wall.
- A false aneurysm may be the result of a prior surgery or trauma.

Aortic Aneurysms



Aortic Aneurysms

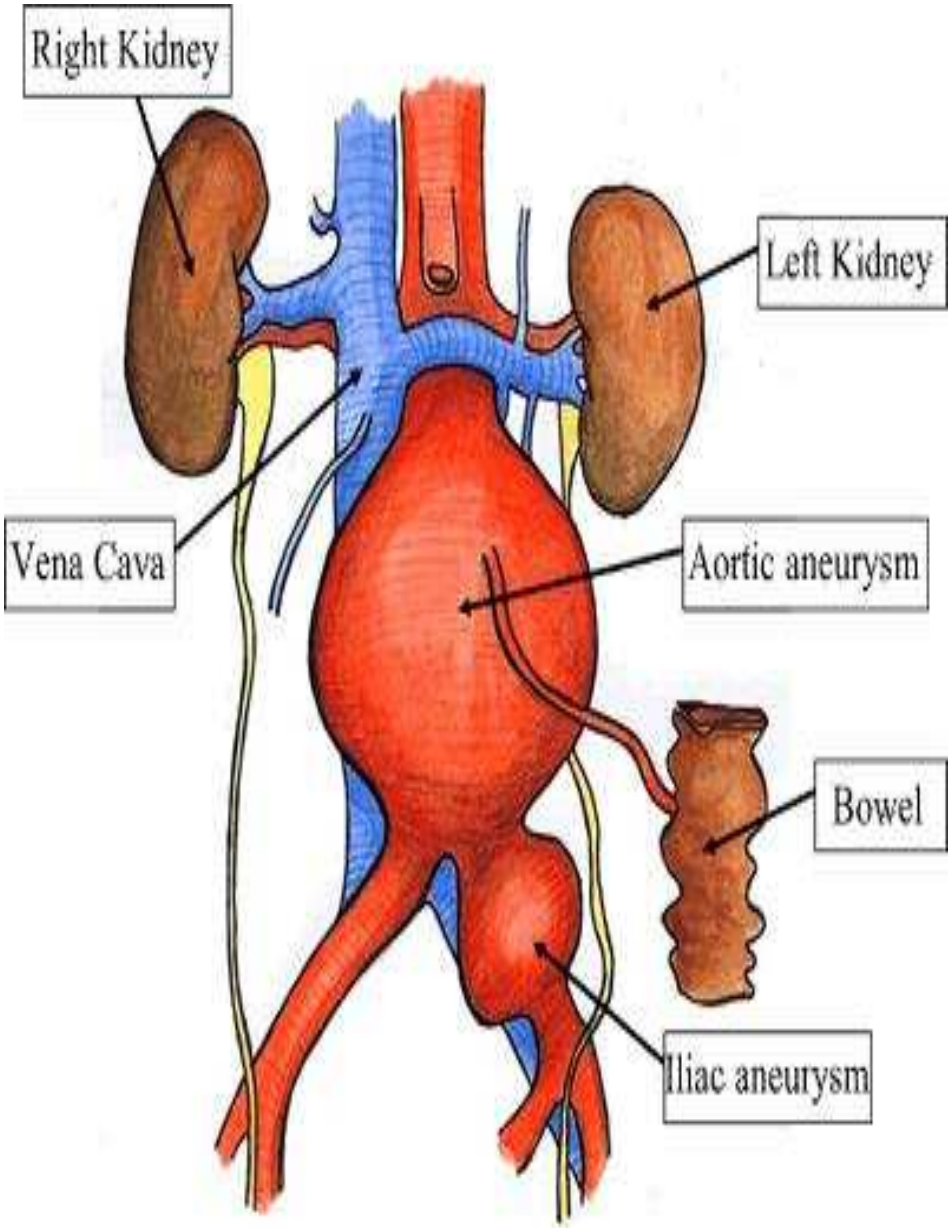
3. Classifications of Aortic Aneurysms according to their location:

- Aneurysms can form in any section of the aorta, but they most commonly in: -

(A) Abdominal Aortic Aneurysms (AAAs): present at the belly area of aorta. Affects men four times more often than women and is most prevalent in elderly patients.

Most of these aneurysms occur below renal arteries (infrarenal aneurysm).If untreated, the eventual outcome may be rupture and death.

Abdominal Aortic Aneurysms



Abdominal Aortic Aneurysms

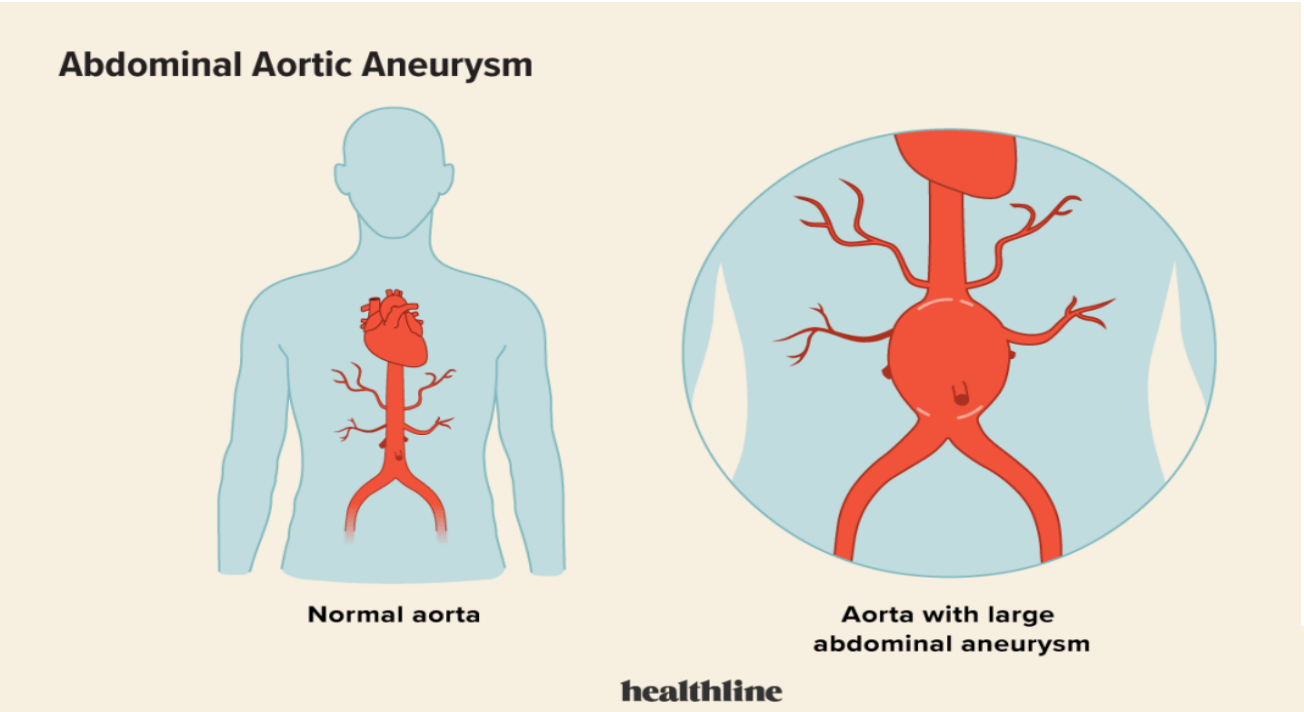
Clinical manifestations Abdominal Aortic Aneurysms (AAAs):

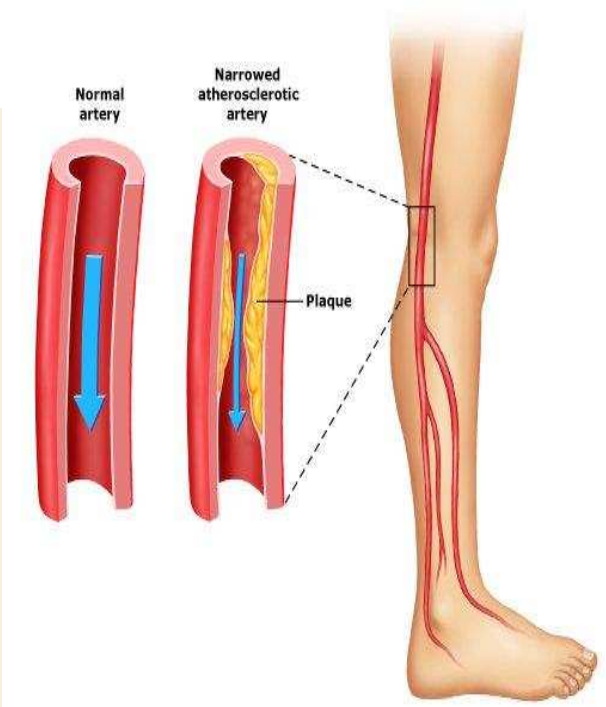
- Patients with (AAA) feel their heart beating in their abdomen when lying down.
- Client´s awareness of a pulsating mass in the abdomen, with or without pain, followed by abdominal pain and back pain.
- Flank pain may be experienced.
- Aortic calcification noted on x-ray.
- Mild to severe mid-abdominal or lumbar back pain.
- Cool, cyanotic extrimities if iliac arteries are involved.

Abdominal Aortic Aneurysms

- Clinical manifestations Abdominal Aortic Aneurysms (AAAs):

  - Claudication (ischemic pain with exercise, relieved by rest).
  - Complication: peripheral emboli to lower extrimities.
  - Rupture and hemorrage.

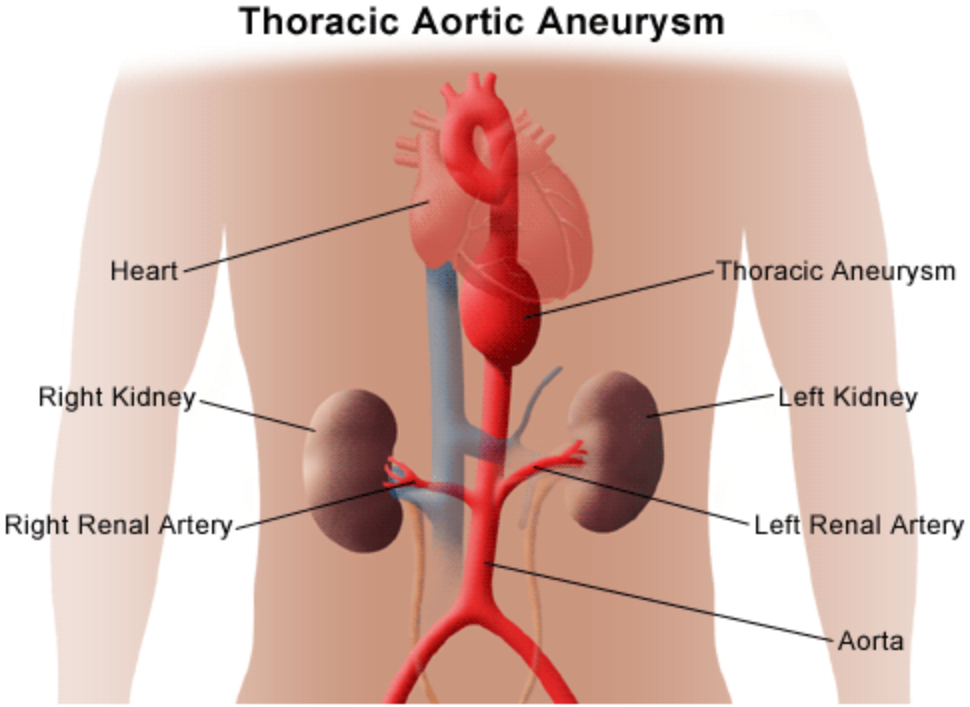




Thoracic Aortic Aneurysms

(B) Thoracic Aortic Aneurysms (TAAs): present at the upper body of aorta. It's also known as ascending or descending aortic aneurysms.

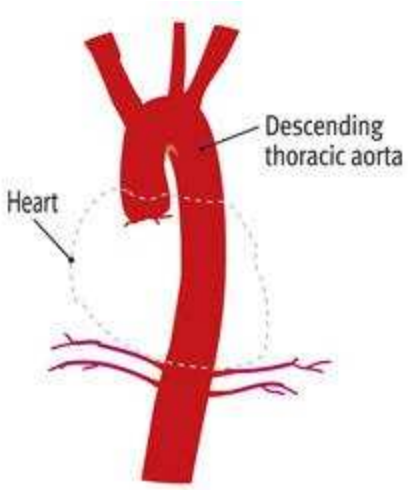
- Occur most frequently in men between the ages 40 and 70 years.
- About one third of patients with (TAA) die of rupture of the aneurysm.



Thoracic Aortic Aneurysms

❑ Clinical manifestations of Thoracic Aortic Aneurysms (TAAs):

- ➢ Back, neck or substernal pain.
- ➢ Dyspnea, and stridor.
- ➢ Hoarseness of speech .
- ➢ Edema of the face and neck.
- ➢ Distended neck vein.
- ➢ Aphonia
- ➢ Dysphagia
- ➢ Complications: such as rupture and hemorrhage.



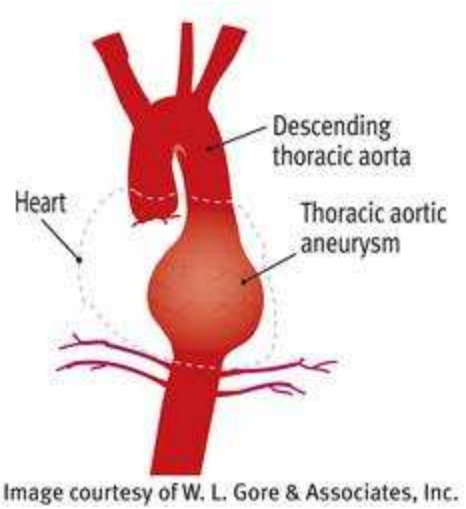
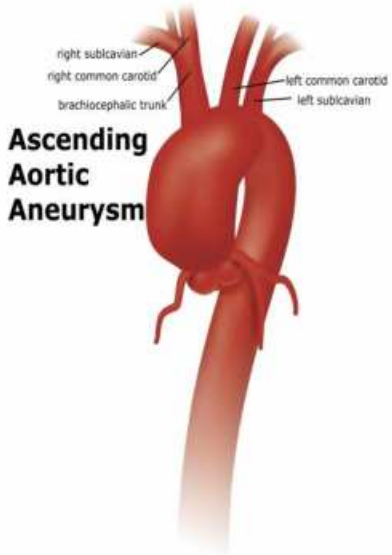


Image courtesy of W. L. Gore & Associates, Inc.



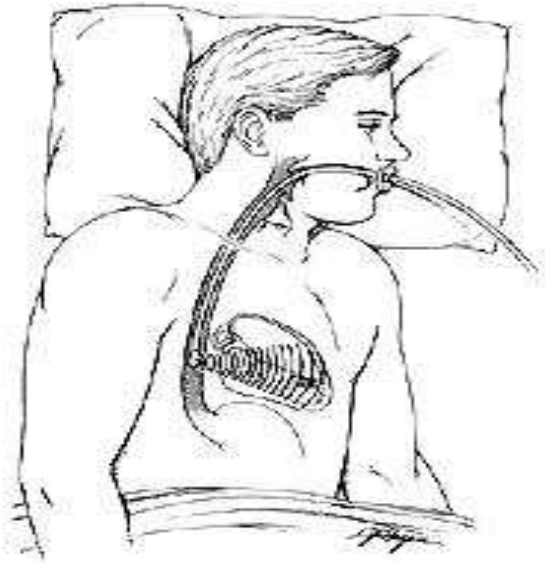
Thoracic Aortic Aneurysms

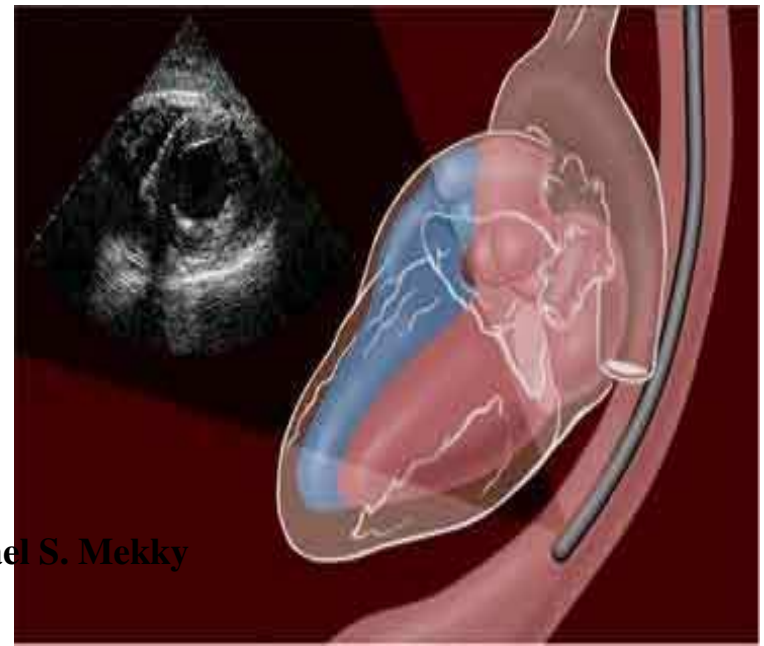
- Diagnostic Tests for Thoracic Aortic Aneurysms (TAAs):

* Chest x-ray

* Computed tomography (CT)

* Transesophageal Echocardiography





Aortic Aneurysms

Nursing Interventions to prevent Aortic Aneurysms

Rupture:

- Maintain bed rest with legs flat.
- Maintain a calm environment, implementing measures to reduce psychologic stress.
- Prevent straining during deafecation.
- Administer beta blockers and antihypertensive as prescibed.
- Elevating or crossing the legs restricts peripheral blood flow and increases pressure in the aorta or iliac arteries.

Aortic Aneurysms

Medical Management for Aortic Aneurysms:

- Control blood pressure: by antihypertensive drugs and controlling risk factors.
- Surgical repair
- Open heart surgery to remove aneurism
- Abdominal surgery is the treatment of choice for abdominal aneurysms wider than 5 cm.

Thank You !